Provider Choice, Home Births, and Health Outcomes[*] [†]

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Updated Version

Abstract

Home births in the US have increased twofold since the early 2000s, yet little is known about the drivers of this increase and its implications for maternal and infant health. In this paper, I use natality data from 1989 to 2021 to estimate relative changes in the prevalence of home birth and subsequent health outcomes before and after an expansion of provider choice. Staggered difference-in-differences estimates indicate a 20–30 percent increase in home births in response to states increasing provider choice by allowing non-nursing midwives to practice through licensing. I find the change in access is most salient for college-educated mothers, low-risk pregnancies, and mothers who pay out of pocket. I find little evidence of a measurable effect of increasing access to midwives on infant health or maternal health.

JEL Classification: I14, I18, J18, J16
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I. Introduction

Rates of home births in the US have doubled in recent years, from 21,000 births occurring at home in 2008 to 45,000 by 2021. Despite the dramatic increase, substantial controversy remains on the safety and implications of home births. The American College of Obstetricians and Gynecologists (ACOG) states that the hospital is the safest place to give birth, and many healthcare professionals argue that home birth is dangerous even for low-risk mothers (AMA, 2021; Sánchez-Redondo, Cernada, Boix, et al., 2020). However, proponents of home birth say if done safely, it can reduce medical interventions and improve the birthing experience for mothers. Moreover, the costs of home births are typically borne by the mothers themselves, and safe low-risk home births could free up hospital beds for others in need.

To better understand the increasing trends in home births and whether these trends yield negative health consequences, I study an expansion of provider choice for birth through state licensing of non-nursing midwives to show that these licensing laws increase the prevalence of home births. I then estimate the effects of the licensing on infant and maternal health outcomes. In particular, I exploit timing in state adoption of licensing laws for non-nurse midwives with a difference-in-differences estimation strategy to estimate the effects of licensing non-nurse midwives on the prevalence of home births in a state and the subsequent health outcomes of mothers and infants.

I focus on state licensing legislation from 1993 to 2019 using natality data from 1989 to 2021. In this period, the majority of states transitioned from not allowing non-nursing midwives to practice to allowing them and requiring licenses.[1] The term "non-nursing midwife" refers to all midwives without a nursing degree. While the exact requirements for licensure differ across states, non-nurse midwives are similar in that they primarily practice in homes, and they are regulated separately from nursing midwives who primarily practice in hospitals.[2] Because states begin licensing non-nursing midwives in different years, I leverage the variation in timing across states.

[1]In some states, prior to licensing, non-nursing midwives were allowed to practice but not formally licensed.

[2]Nurse midwives are already regulated in each state during this period, and only 7 percent of nurse midwives report practicing in homes (GAO, 2023).

First, I show that when a state begins licensing non-nurse midwives, there is an increase in the number of practicing certified professional midwives (CPMs), a subset of non-nurse midwives. Because non-nurse midwives are the most common attendants at home births, I then study the effect of the legislation on the prevalence of home births using birth records from the National Center for Health Statistics from 1989 to 2021. I show that the 20-30 percent increase in home births is driven by a positively selected group of college-educated and low-risk mothers who pay out of pocket. This corresponds to around 4,000 additional home births per year. While there is a substantial increase in home births, I find little evidence of measurable negative effects on infant and maternal health. In particular, I find no statistically significant effect on prenatal visits, low birth weight, or premature birth. My sample is large enough to detect effects of midwifery licensing laws for small changes in these outcomes.[3] However, midwifery licensing laws reduce inductions by 3.6 percentage points.

Because birth data do not allow direct observation of mothers who are transferred to a hospital during labor from an intended home birth, I use hospital discharge data from the Healthcare Cost and Utilization Project (HCUP) to study hospitalizations of mothers and infants due to delivery complications. I find no evidence of increased hospitalizations for mothers or infants after non-nurse midwifery legislation. Overall, the difference-in-differences results suggest that increasing access to non-nurse midwives can be cost saving for society by increasing home births for low-risk pregnancies with limited negative health effects and be welfare improving for a mother by increasing options for the delivery place, as evidence by their revealed preferences. However, because I cannot directly test the causal impact of home births on health, the overall implications of a larger shift in home births remain an open question.

Mothers' existing formal healthcare is likely to influence their response to the increase in provider choice. I show that the initial choice set matters, and women living in rural counties increase their take-up of home births more than women living in urban counties. Access to maternity care differs vastly between rural and urban areas in the US, with 2.2 million women

[3] I present some evidence of an increase in neonatal seizures, which is a rare outcome that occurs when infants lose access to oxygen. However, this estimate is not robust to multiple hypothesis testing.

living in counties defined as maternity care deserts with limited access to prenatal care (MOD, 2022). However, because the women who increase home birth after midwifery laws are low-risk and have higher socioeconomic status (SES) as proxied by educational attainment, they are less likely to be constrained by the limited access to obstetric care prior to midwifery licensing than other rural women who are not college-educated and have lower socioeconomic status. While many proponents of midwifery laws cite them as the solution to maternity care deserts for all women, I present some evidence that, in fact, it is the least vulnerable that shift their behavior when midwifery laws are passed.

My findings contribute to the existing literature in several ways. I present new evidence on how modern midwifery laws affect the prevalence of home births and health outcomes. Previous literature on midwives in economics examines the health impact of changes in the scope of practice for nurse midwives on hospital births.[4] Markowitz, Adams, Lewitt, and Dunlop (2017) and Hoehn-Velasco, Jolles, Plemmons, and Silverio-Murillo (2023) show that increases in the scope of practice for nurse midwives results in little change in health outcomes but does decrease cesareans. I build on these studies by studying the effect of licensing that allows non-nurse midwives to begin practicing on the prevalence of home births. Hoehn-Velasco, Jolles, Plemmons, and Silverio-Murillo (2023) present results showing that there is a negative effect of CPM licensing on Certified *Nurse* Midwife (CNM) use, suggesting a level of substitutability between the two. Alongside previous work my findings of limited health effects of allowing non-nurse midwives to practice suggest that both nurse and non-nurse midwives may serve as relatively safe and less costly alternatives to traditional OB-led births.

Other existing studies look at midwives in a historical context. Using early midwifery regulation from 1900–1940, Anderson, Brown, Charles, and Rees (2020) find that requiring midwives to be licensed reduces maternal mortality by 6 to 7 percent. Further, these early midwifery regulations affect long-run outcomes by increasing longevity and reducing long-run mortality of children (Fletcher and Noghanibehambari, 2023). Increasing the availability of trained midwives in Sweden

[4]From 1989 to 2021, nurse midwives were licensed in all US states; however, there is variation in the scope of practice for these midwives.

in the 19th century also reduced maternal mortality (Lorentzon and Pettersson-Lidbom, 2021). My paper presents new estimates on non-nurse midwifery licensing in the modern context. While the historical research on midwives informs us about the general effects of licensing laws, it cannot speak to the effect of increasing provider choice and access to home birth in the modern context where birth has been medicalized, and medical technology has changed vastly (Goode and Rothman, 2017).

Because home birth in the US is relatively uncommon, it has been studied far less than in European contexts where it occurs more regularly in both historic and modern settings. Particularly, Daysal, Trandafir, and van Ewijk (2015) finds that in the context of the Netherlands, giving birth at a hospital instead of at home reduces infant mortality likely due to care after birth. Moreover, openings of maternity wards in Sweden in the 1900s reduced home deliveries with midwives and reduced neonatal mortality (Lazuka, 2023). These papers motivate the policy importance of studying the rise in home births in the US, a setting without universal healthcare, and the potential impacts of the move to midwifery care on health outcomes.

More broadly, my findings contribute to the literature on behavioral responses to provider choice and the heterogeneity of impacts for different populations. Much of the existing literature focuses on physical access to care, showing that distance to the nearest hospital affects take-up of preventative care, especially for Black children (Currie and Reagan, 2003). For pregnant women in particular, Aizer, Lleras-Muney, and Stabile (2005) show that Black mothers respond less to an increase in provider choice than White mothers. In contrast to the previous literature, I find that both White and Black mothers respond to increases in provider choice, with heterogeneity being driven by education level.

Finally, I study the health effects of a change in provider choice, contributing to a growing literature on the effects of maternity ward closures in the US on health. Research shows closures lead to a reduction in cesareans and little impact on infant health for mothers living in the most rural areas (Fischer, Royer, and White, 2024; Battaglia, 2022; Durrance, Guldi, and Schulkind, 2024). I contribute by studying the health impacts of an increase in provider choice for prenatal care and delivery using midwifery laws which expand access instead of reducing access.

II. Background

II.1. Non-Nurse Midwifery Licensing

Midwives have a long history of performing home births. As Goode and Rothman (2017) explain, midwifery in the US is inextricably linked with enslaved people and began in the South with "grand" midwives who were enslaved and attended births for other enslaved people and plantation owners. They were often not recognized as midwives until they witnessed, attended, and supervised many births. Before the 1930s and the introduction of sulfa drugs, giving birth in hospitals did not reduce maternal mortality, and the majority of births occurred at home (Jayachandran, Lleras-Muney, and Smith, 2010). Now, around 1 percent of all births in the US are home births, most often performed by a non-nursing midwife.

There are two main categories of midwifery – nurse and non-nurse.[5] Nurse midwives are registered nurses who have a master's degree in midwifery and primarily practice in hospitals and birthing centers. Each state in the US regulates nurse midwives, often called Certified Nurse Midwives (CNMs). Non-nurse midwives differ from nurse midwives in that they do not have a nursing degree; however, within the category of non-nurse midwives, there are different types of licenses. The three most common types of non-nurse midwives are Certified Midwives (CMs), Certified Professional Midwives (CPMs), and direct entry or licensed midwives. CMs are closely related to CNMs because they have a master's degree and are licensed by the American Midwifery Certification Board, which also licenses CNMs (GAO, 2023). Given this relation, I have excluded states which license CMs but not any additional type of non-nurse midwife from the main analysis.[6] Aside from CMs, I group all other non-nurse midwife licensing legislation together for a number of reasons. Firstly, licensing of CPMs and direct-entry midwives often happens in tandem, or the

[5]In contrast to midwives, Doulas do not perform medical interventions. However, they can work in conjunction with a doctor or midwife.

[6]Only New Hampshire and New York license CMs and no other type of non-nurse midwife, so they are excluded. I show robustness to including these states in Section V.2

wording of legislation only refers to "non-nurse midwives". Additionally, while the exact terms used in each law may differ, there is an explicit difference between nurse and non-nurse midwives and where each primarily practices, hospitals and homes, respectively.[7]

Of non-nurse midwives, CPM is the most common type of midwife that states license. CPMs are not required to have an academic degree but do have supervised clinical requirements and are certified by the North American Registry of Midwives. The clinical education must last 2 years and include a minimum of 55 births NARM (2025). Midwives referred to as direct entry, licensed, or lay midwives are not overseen and licensed by a national board; rather, each state determines its own licensing requirements. Regardless of type, non-nurse midwives can provide comprehensive care throughout the pregnancy, birth, and postpartum period. Specifically, they can conduct all prenatal visits in the pregnant woman's home and order ultrasounds and other necessary labs. They are trained to recognize and refer high-risk pregnancies to obstetricians. While they do not have prescriptive authority, they can administer medications and use devices based on state-specific regulations but are trained only for vaginal deliveries and do not practice in hospital settings (Vedam, Stoll, MacDorman, Declercq, Cramer, Cheyney, Fisher, Butt, Yang, and Kennedy, 2018).

Prior to the licensing of non-nurse midwives within a state, home births could occur, performed by physicians, nurse midwives, or unlicensed (and often illegal) non-nurse midwives. However, home births with physicians and nurse midwives are uncommon. In fact, only 7 percent of all nurse midwives report practicing in home birth settings (GAO, 2023), and in 2010, only 2.5 percent of all home births were attended by doctors. After the licensing of non-nurse midwives, home births could be performed by physicians, nurse midwives, and non-nurse midwives.

Lastly, I explore *why* states have adopted the licensing laws. The reasons can be categorized into 3 broad categories. Firstly, policy makers realize that home births are occurring with unlicensed or illegal midwives, and they desire to make those births safer. This is often a result of a publicized story of a midwife being arrested after the death of an infant, as occurred in Oregon in 2013 (Tobias,

[7] The appendix contains details on the legislation in each state.

2013). Secondly, in some states, midwives have been pushing grassroots campaigns to allow the licensing of midwives for years. For example, when the midwifery law in Missouri passed in 2007, midwives had been fighting for it since the 1990s (Eberlin, 2008). Finally, as attention on maternity care desserts has grown, so has the desire to provide healthcare to rural women. Thus, states like Kentucky have allowed licensing to fill gaps in healthcare in rural areas (Close, 2022).

Given the motivations for licensing non-nurse midwives, it is not ex ante clear how home births and health outcomes will change. If home births occur despite midwives being unlicensed, enhanced training through licensing legislation for those midwives would likely result in little change in the number of home births but significant improvements in health outcomes. Alternatively, if midwives were not practicing prior to state licensing laws, we should expect home births to increase after they begin practicing with an ambiguous impact on health given the unknown impact of home birth on health. Relatedly, I explore the differences in the extent to which non-nurse midwives were restricted from working before a state's licensing law. In some states, midwives found practicing before the licensing laws were arrested. However, in other states, midwives were left to practice unlicensed freely. I explore the heterogeneity in the strictness of midwifery allowance prior to legislation in heterogeneity analysis in Section V.1.2.

III. Data

As of 2021, 33 states allow and license non-nurse midwives. I have collected data from state legislative records to establish a data set of licensing timing. The treatment year is defined as the year when a state passed legislation allowing the licensing of non-nurse midwives. Although the difference between passage and effective start dates of licensing varies across states, all states begin licensing non-nurse midwives in the following year at the latest.[8] New York and New Hampshire are excluded because they solely license CMs, which are more closely related to CNMs than other types of non-nursing midwives. I drop Hawaii, Kansas, Maine, Nevada, and West Virginia in the

[8] In the case of Missouri, in 2007, the state legalized non-nursing midwifery, but it was delayed in court. In 2008, the Missouri Supreme Court upheld the law. In this case, I use 2008 as the treatment year.

main analysis. In these states, midwifery was not licensed during the time period of this study, but they do always allow midwives to practice because birth is considered a natural occurrence instead of medical.[9] Lastly, I exclude Delaware due to conflicting reports of years of licensing legislation.

Figure 1 shows how states have adopted midwifery licensing over time.[10] I present summary statistics in Table 1, showing that states which pass midwifery licensing legislation and states that do not pass legislation by 2021 have similar birth rates and populations of birthing-aged females, nonwhite populations, and economic measures. Overall, there are 8 treated before 1989, 23 states included as eventually treated, and 11 states are never treated.

I use two datasets to measure the supply of midwives. Firstly, I use data on the number of CNMs in each state from The Nurse Practitioner's annual legislative update to identify nurse midwives, which publishes counts of licensed CNM from 2000–2021. Then, to identify a measure of non-nurse midwives, I use data on CPM from The North American Registry of Midwives (NARM) annual counts of CPM in each state from 2001 to 2020. This data is missing for 2011, 2017, and 2018, so the results must be interpreted without these years. In addition, data on direct entry non-nurse midwives is not reported or available on a consistent basis. Therefore the measure of supply of non-nurse midwives is necessarily an under count as it only includes CPMs and misses all other non-nurse midwives.

I use the National Center of Health Statistics Vital Statistics birth records from 1989 to 2021 to analyze the effect of this increase in provider choice on birthplace (NCHS, 2022). These birth certificate data contain demographic data on the mother, including race, age, educational attainment, and payment source.[11] They also contain the place of birth and county of residence. I categorize place of birth as a home or non-home birth.[12] In the 2003 revision of the birth records, states began recording for home births only, whether it was planned or unplanned. For consistency of coding before and after the revision, I define home birth as any birth that occurred at home regardless of it

[9] In subsequent analysis, I show that estimates are robust to including these states in the comparison group and including New York and New Hampshire as treated states.

[10] The appendix contains details on the legislation in each state.

[11] Data on the payment source is only available from 2009-2021.

[12] I also use hospital or non-hospital births, and estimates are consistent across non-hospital and home births.

was planned.[13] Notably, if the birth was intended to take place at home, but resulted in a transfer and birth at the hospital, it is reported as a hospital birth. Figure 2 shows the twofold increase in home births in the US from 1989 to 2021, an increase from around 21,000 to 45,000 births in a given year. In addition to information on the birth location, these data also have information on delivery, such as the delivery method and inductions. They also have data on maternal health characteristics, which I use to define high-risk pregnancies and low-risk pregnancies. Births are defined as high-risk if one or more of the following conditions are met: breech, hypertension (chronic or pregnancy-related), eclampsia, premature, previous cesarean, or a plural birth. Births are low-risk when none of the former are true.

If the expansion of provider choice increases home births, it may also affect the health of infants and mothers or change the composition of infants born outside of a hospital. To measure infant health, I use the birth certificate data, which contains information about the infant's health at birth.[14] Because midwives provide prenatal care and perform deliveries, I examine outcomes that would be affected by prenatal care quality such as birth weight and outcomes that are unanticipated but can be a sign of delivery complications such as neonatal seizures. While most of my outcomes use birth certificate records, I also provide estimates on fetal and infant death with data from the Center for Disease Control (CDC) Wonder from 2005 to 2021. Because perinatal deaths are 1 to 2 times more likely to occur in planned home births than in the hospital (ACOG, 2017), I focus on fetal deaths as a proxy for complications prior to birth and infant deaths within 24 hours of birth as a proxy for post-labor complications.[15] Finally, I use the National Immunization Survey to see if there is a reduction in Hepatitis B vaccinations at birth (NIS, 2023). Vaccination could change if switching to home births disrupts or changes the connection between families and formal healthcare. The first dose of Hepatitis B vaccination is recommended to be given to an infant within 24 hours of

[13]For the subset of states that had revised their birth records by 2010, the rate of unintended home births from 2010-2021 is only 0.13 percent.

[14]It is possible that midwives report health outcomes for home births on the birth records systematically different than hospital births. To determine if differential reporting is likely occurring, I estimate the effect of midwifery legislation on reporting birth weights at round numbers. I find no change in reporting birthweight at 1000 grams, 1500 grams, 2500 grams, 3000 grams, or 3500 grams.

[15]I also provide estimates for abortion using KFF state and year 2009-2020 abortion data to rule out changes in the likelihood of giving birth.

birth. In all but one state that regulates non-nurse midwives, midwives are not licensed to give this vaccination.

Measuring maternal health with birth record data is challenging because birth data do not allow direct observation of mothers who are transferred to a hospital during labor from an intended home birth. To estimate the effects on overall maternal health and determine if mothers or their infants are being transferred to the hospital during or within a day of delivery, I use data from National Inpatient Hospitalization from The Healthcare Cost and Utilization Project (HCUP) (HCUP, 2011). These data include 100 percent of hospital discharges from a sample of hospitals in each state from 1998 to 2011.

Finally, I use Area Health Resource Files (AHRF) to provide additional insight into the places where midwifery licensing legislation is most salient (USDA, 2024). These data contain rural-urban continuum codes and data on hospital access for each county in the US. I use population data from the National Cancer Institute's Surveillance, Epidemiology, and End Results Program (SEER) to add county and state-level covariates for the total population and the Current Population Survey (CPS) for state-level unemployment, poverty rates, and median household income (SEER, 2023; Flood, King, Rodgers, Ruggles, Warren, Backman, Chen, Cooper, Richards, Schouweiler, and Westberry, 2024).

IV. Empirical Approach

My preferred estimation strategy for studying the effect of an increase in provider choice is a difference-in-differences design that exploits the variation in state timing of non-nurse midwife licensing legislation. The variation in treatment timing could result in biased estimates from a TWFE design if treatment effects are dynamic (Goodman-Bacon, 2021).[16] To avoid biased estimates, I use methods proposed in Callaway and Sant'Anna (2021) that compare changes in birth-related outcomes of states that pass non-nurse midwifery licensing legislation relative to

[16]Negative weights from TWFE occur when already treated states are used as a comparison for currently treated states. Because it may take a few years for the demand for home birth to change after midwives are allowed to practice treatment, effects are likely dynamic in this setting.

those that have not yet passed such legislation or never do so. Because the data begins in 1989, I define states treated in 1989 or earlier as always treated. I balance the panel to include treated states with 4 years of data prior to treatment and 4 years after. Thus, there are 18 treated states from 1993 to 2017. Formally, to estimate the average treatment effect on the treated (ATT), I use weighted least squares. I assume that there is no treatment anticipation $\delta = 0$ and parallel trends between group g and groups that are "not yet treated" by time t to identify ATT(g, t):

$$ATT(g,t) = E[Y_t - Y_{g-1}|G_g = 1] - E[Y_t - Y_{g-1}|D_t = 0] \tag{1}$$

Y_t is the outcome of interest such as rate of home birth and infant health measures at time t, G_g is a binary variable equaling one if the state is first treated in period g, and D_t is a binary variable equaling zero for states "not yet treated" by t.

Aggregating the group time average treatment effects together gives the overall effect of midwifery licensing on place of birth and health. This aggregation can be done in multiple ways, and I present results using 3 methods. The first is "Simple Aggregation," which is the weighted average of all group-time average treatment effects based on group size. This method puts more weight on ATT(g,t)'s with larger group sizes and observations further past treatment. Next is "Event Time Aggregation," where I aggregate across event time. This aggregation is used in event study figures and gives the average effect of participating over specific years of exposure. Lastly, the "Group Time Aggregation" first computes the average of treatment for each group time and then averages those effects together. This creates a measure of the average effect of treatment experienced by all states ever treated.[17] I weight states by the number of births to improve efficiency and cluster standard errors at the state level. The key identifying assumption is that in the absence of a change in provider choice, trends in home birth and health outcomes would have remained parallel in treatment and comparison states. A main corollary of this assumption is that demand for home births did not induce states to begin licensing non-nurse midwives. In the main specification the reference

[17] The main tables report "Simple Aggregation", and the main figures show "Event Aggregation" but I show robustness to "Group Aggregation" in the appendix Tables A3 and A4.

period is the the period prior to treatment for both the pre-and post-treatment coefficients as is traditionally done in TWFE. I also show the event study estimates using "pseudo-ATT estimates" where the reference period for the pre-treatment coefficients is the period immediately preceding it in Figure A5.

Before I discuss the estimated effects of increasing provider choice through the licensing of non-nurse midwives, I present evidence to support the identifying assumption. Table 2 shows that time-varying state-level maternal characteristics, median household income, CNM scope of practice laws, and state adoption of Perinatal Quality Collaboratives are uncorrelated with the timing of state midwifery licensing legislation.[18] States are not more likely to adopt other programs aimed at improving the healths of infants at the same time as midwifery licensing. These results show that it is unlikely states opt into non-nurse midwifery licensing because of a broader trend in preferences for nurse midwives or concern for maternal and infant health. Further, treatment is uncorrelated with the baseline rates of home births, inductions, or midwife-attended births. This result, shown in Table 3, suggests that baseline preferences for home birth and health are not driving the uptake of legislation. Moreover, my event study figures in Section V provide visual evidence that treatment is uncorrelated with existing trends in preferences for home birth and health outcomes. Another threat to identification occurs if maternity ward closures induce midwifery legislation. Figure 3 shows that maternity ward closures do not cause midwifery licensing laws.[19]

V. Results

First, I study the impact of midwifery licensing legislation on the number of CPMs within a state. Then, I estimate the effects of midwifery legislation on the percentage of home births in a state and infant and maternal health using NCHS birth records data and HCUP hospitalization data. Finally, I show robustness to weighting and comparison group as well as the choice of estimator in Section V.2.

[18] Perinatal Quality Collaboratives are state-level programs aimed at improving maternal and infant health.

[19] To identify a maternity ward closure, I use the method proposed in Fischer, Royer, and White (2024) where a county is defined as having a closure if in year n a hospital has 75 percent more hospital births than it does in year n+1.

V.1. Effect of Midwifery Licensing

V.1.1. Evidence of an increase in midwives

Estimates indicate when a state begins licensing non-nurse midwives to practice, the number of licensed CPMs increases.[20] In panel (a) of Figure 4, I find an increase in the supply of CPMs practicing in a state when states begin allowing non-nurse midwives, suggesting there is a take-up of non-nursing midwifery as a career. Furthermore, panel (b) of Figure 4 shows that the supply of nurse midwives decreases by a small amount, suggesting some level of substitution between the two careers. Hoehn-Velasco, Jolles, Plemmons, and Silverio-Murillo (2023) also find evidence of this substitution when looking directly at the impact of CPM licensing legislation on CNM use. However, the measure of supply non-nurse midwives only includes CPM, a subset of non-nurse midwives. Because I study all non-nurse midwifery legislation which includes legislation for CPMs and all other non-nurse midwives including direct entry midwives for which there is no national data source, I am under counting the supply of non-nurse midwives. Despite this limitation, Figure 4 provides evidence that more of one type of non-nurse midwives, CPMs, practice when they are allowed to be licensed.

V.1.2. Changes in place of birth

In Figure 5, I present graphs plotting the event study estimates for the effect of licensing on the percent of home birth and their corresponding 95% confidence intervals from the WLS model in Equation (1), controlling for state and year fixed effects. First, I note that prior to event time zero, when a state passes midwifery legislation, the estimates are statistically similar to zero, suggesting that prior to state licensing, outcomes in treatment and comparison states trend in parallel. When a state begins licensing midwives, estimates indicate a 0.21 percentage point increase in the share of births occurring at home, a 31 percent increase relative to the baseline mean. Table 4 column

[20]In many states, it was illegal to practice as a non-nurse midwife before licensing legislation; however, in some states, it was allowed but unlicensed. For this reason, the reporting of CPMs in the pre-period is non-zero.

1 reports the corresponding simple ATT estimate.[21] Table 4 also shows that results are robust to unweighted estimates. Because non-nurse midwives also practice in birthing facilities, I estimate the effect of the midwifery legislation on births in birthing centers directly in Figure 6. I find no significant effect of midwifery legislation on the rate of births occurring in birthing facilities, suggesting that that legislation most directly impacts home births as opposed to all out of hospital births.

Advocates for home birth argue that it is safe for low-risk pregnancies. Therefore, if home births increasingly involve high-risk mothers instead, even home-birth advocates would argue against them. Thus, exploring what types of mothers drive the increase in home births is particularly important. Figure 7 shows the simple aggregation coefficients and 95% confidence intervals for different subgroups while their corresponding event studies are shown in Figure 8. More formally, Columns 2-11 of Table 5 show the ATT estimates on the percent of home births for each subgroup. I find that the increase in home births is driven by college-educated mothers and those with "low-risk" pregnancies. Low-risk pregnancies are singleton pregnancies in which the mother does not have hypertension or eclampsia, has not had a previous cesarean, and the infant is not breech or premature. In particular, for low-risk mothers, home births increased by 30 percent. I also show that home births increase by 29 percent for college educated mothers, and there is no statistically significant change for non-college educated mothers.[22] This heterogeneity analysis suggests that when a state increases access to home birth through non-nurse midwives, positively selected mothers shift into home births. In addition, while White mothers have a higher baseline likelihood of giving birth at home, non-nurse midwife licensing causes a 64 percent increase in home births for Black mothers and only a 20 percent increase for White mothers. The large increase in home births for Black mothers is likely related to their experience giving birth in hospitals and the large

[21] While the main outcome of interest is the percent of home births, the Natality data also includes information on the reported attendant at birth. Figure A1 shows event study estimates for the effect of midwifery licensing on the percent of births attended by a non-nurse midwife. While less precise, there is an increase in births attended by non-nurse midwives. Research suggest that data on attendant at birth is often misreported or missing (Faucett and Kennedy, 2020); in fact, in 2010, 25 percent of home births had the attendant reported as "other" or missing.

[22] From 2003-2013 there is inconsitency across states in reporting of education resulting in states with low reporting of the college education variable. In Table A1 I show robustness to excluding states that have missing data on education for more than 5 percent of the births in a given year.

gap in maternal morbidities and mortality between Black and White mothers. Black mothers are increasingly looking for options outside of the traditional hospital birth (Reissig, Fair, Houpt, and Latham, 2021); (Suarez, 2020). Further, as Figure 9 shows, the increase is driven by births paid for out of pocket. I discuss the implications of the majority of births being paid out of pocket in further detail in Section VI.

In addition to the heterogeneity across characteristics of mothers, there may also be important differences in the effect of midwifery licensing based on the availability of other healthcare. Access to maternity care differs vastly between rural and urban areas in the US, with over 2 million women living in counties defined as maternity care deserts with limited access to prenatal care (MOD, 2022). To investigate to what extent healthcare access plays a role in inducing home births after midwifery licensing legislation, I estimate the effects of Equation (1) for metro and non-metro counties. In Figure 10, I present graphs plotting the event time aggregations corresponding to Table 6. Panel (a) shows that non-metro counties rather than metro counties drive the effect on home births. Rural counties have lower access to healthcare, so these results provide evidence that the licensing legislation of midwives is particularly salient in places with less access to standard healthcare.

States differ vastly on several factors related to the likelihood that someone may switch to home birth. Firstly, some states provide Medicaid coverage of home birth while others do not. Table 7 columns 1 and 2 explore this heterogeneity.[23] There is no detectable heterogeneous effect on home births regardless of state Medicaid coverage of it, likely due to a loss of power. Next, I estimate the effect on home births for states that have high and low maternal mortality.[24] Again, Columns 3 and 4 show that there is no distinguishable differential effect for states that have above average maternal mortality. Lastly, to investigate if the magnitude of the effect of licensing legislation on home births differs by the pre-period strictness of state laws regarding midwives, I separate effects on home birth by "strict" and "non-strict" states.[25] Columns 5 and 6 show estimates of home birth by

[23]The states that have no Medicaid coverage are: AL, CO, ID, IN, MD, MI, MO, SD, TN, UT, and WI

[24]The states categorized as low maternal mortality are defined by having maternal mortality below the national average from 2018–2021: AK, CA, CO, ID, MD, MI, MN, MT, OR, RI, SD, UT, VT, WA, WI, and WY.

[25]In the analysis for strict states I exclude states that explicitly were lenient on midwives or allowed optional licensing

state strictness prior. Point estimates suggest a larger increase in home births in states with stricter pre-legislation treatment of non-nurse midwives, consistent with licensure legislation relaxing a more binding constraint. However, these estimates are not statistically different than one another. Still, this finding fits with ex-ante expectations that there will be a smaller effect on home births in states that were more friendly to home birth prior to licensing legislation.

V.1.3. Delivery and Infant Health Effects

Next, I estimate the effect of non-nursing midwifery licensing legislation on delivery outcomes. Delivery outcomes, which may differ by place of birth, include delivery method, if the delivery was early (37 or 38 weeks gestation), and inductions. Women self-select into home birth often due to a desire for fewer medical interventions. Because midwives cannot perform cesareans and, in most states, are not allowed to administer medication that induces delivery, a reduction in these outcomes could be expected from an increase in home births. I find no measurable effect on cesareans or forceps usage as seen in Table 8. However, I find a reduction in inductions of 3.6 percentage points, corresponding to a 16 percent change from the mean. Because the reduction in inductions is larger than the increase in home births, some of the reduction in inductions must come from hospital births. An explanation for this discrepancy is that some non-nurse midwives offer hybrid care where the midwife does all prenatal care and begins labor at home, but the actual delivery occurs at the hospital (MBH, 2021). Additionally, the increase in non-nurse midwives could reduce inductions in hospitals through a reduction in overcrowding and resulting inductions from a need to free up beds, or a change in norms and expectations from patients regarding medical interventions. In Figure 11, I present graphical evidence of the decrease in inductions. Lastly, I find no evidence of a statistically significant change in the number of births as shown in Table 8, which suggests there is not a shift in childbearing decisions.[26]

prior to officially licensing non-nurse midwives. These states include OR, TN, MI, ID, MN, OK, and WI. In the analysis for non-strict states, I exclude states where midwives practicing illegally were arrested, or the midwives were explicitly illegal. These states include IN, SD, VA, MO, CO, MD, and MT.

[26] I also estimate the effects of midwifery licensing on abortions to understand the effect on childbearing decisions and find no evidence of a change in rate of abortions shown in Figure A2.

Although the delivery location is unlikely to affect health outcomes such as birth weight and gestation because low-risk pregnancies drive the increase in home births, non-nurse midwife licensing may still affect these health outcomes by changing the quality or quantity of prenatal care (Corman, Dave, and Reichman, 2019). Table 9 shows simple ATT estimates on birth weight, the probability that an infant is born with low or very low birth weight, and five-minute APGAR score.[27] I find no evidence of an effect of midwifery licensing on these infant health outcomes, other than a small increase in APGAR score which is not robust to multiple hypothesis testing. Finally, consistent with the result of no statistically significant change in birth weight, I find no change in the total number of prenatal visits as shown in Table 9 column (5). In Section V.2. I provide a discussion of power limitations.

Given that the change in home births is driven by low-risk mothers, I also estimate health effects by risk groups and find similar effects among low and high-risk mothers as the main sample for all for infant and delivery outcomes shown in Table 10. The reduction in inductions for both low and high-risk mothers provides evidence that the reduction is likely due to a combination more home births and spillovers to hospital births through a reduction in overcrowding or a change in norms.

Next, I estimate the effects of midwifery licensing on infant health outcomes that are indicative of complications with delivery and are unanticipated: assisted ventilation, neonatal ICU admissions, seizures, and delta APGAR for a subset of states from 2010-2021.[28][29] In Table 11, I show the estimated effects of midwifery licensing on assisted ventilation, NICU admission, neonatal seizures, and delta APGAR for all infants. I show no measurable effect on ventilation, NICU admissions, or delta APGAR. I do find an increase in neonatal seizures of 0.01 percentage points or 20 percent. Notably, neonatal seizures are around two times more likely to happen during a home birth at baseline and are an extremely rare outcome, only occurring in 0.05 percent of all births. Moreover,

[27]Low birth weight is an indicator variable equaling one if the birth weight is smaller than 2500 grams and very low birth weight if smaller than 1500 grams. APGAR score measures health from 1 to 10 taken 5 minutes after birth.

[28]Variables indicating assisted ventilation, NICU admission, and neonatal seizures only became available with the 2003 change to birth certificate. However, because states rolled these changes out over time the only treated states included in Table 11 are IN, MD, MI, OR, and SD.

[29]Delta APGAR is the difference between the 10 minute and 5 minute APGAR scores and can reflect rapid decline of infant health immediately after birth.

as Table 12 shows the increase in neonatal seizures is driven by low-isk mothers who are the most likely to switch into home birth. The full sample increase corresponds to around 300 additional births with neonatal seizures, assuming around four million births in a given year.[30]

Because vital statistics birth data do not allow direct observation of mothers who are transferred to a hospital during labor from an intended home birth, it is possible that some births with negative outcomes are recorded as hospital births but began as home births.[31] If this were to be the case, I would expect to see changes to health across in-hospital births as well. Because I estimate the effect of midwifery licensing on outcomes across all births regardless of where they occur, reduced form estimates suggest this is not occurring. To address the concern that home births are being transferred to the hospital after delivery for emergencies, I use HCUP hospital discharge data to estimate the effects of midwifery licensing on hospital discharges of infants 1 to 7 days old. Table 13 shows that while the estimates are imprecise, I can rule out increases in hospitalization greater than 3.6 per 1000 for infants 1 to 7 days old with 95 percent confidence. While I cannot directly tell if the birth occurred at home, I can estimate the hospitalizations for infants born prior to admission in Column 2. In this case, I can rule out increases greater than 0.025 per 1000 births.

Despite the lack of measurable effect on most reported measures of infant health, I also estimate the effect on the most severe outcome, fetal and infant deaths. Figure 12 shows that there is no measurable effect of midwifery licensing on either infant or fetal death rates. I can rule out effects greater than an increase of 0.33 infant deaths per 1000 births and 0.39 fetal deaths per 1000 births. For comparison, Kennedy-Moulton, Miller, Persson, Rossin-Slater, Wherry, and Aldana (2022) find that fetal death for mothers at the top of the income distribution is 3.4 deaths per 1,000 births and 7.1 for the bottom of the distribution.

In addition to the immediate health effects on infants, there could be longer-run effects on the child's health if having a home birth causes greater detachment from formal medical care. For example, it's recommended that the Hepatitis B vaccination be given to an infant within 24 hours of

[30] However, I note that the effects on neonatal seizures only come from a subset of states and the results are not robust to multiple hypothesis testing.

[31] According to a systematic review 0–5.4 percent of all home births end in an emergency trip to the hospital (Blix, Kumle, Hanne Kjærgaard, and Lindgren, 2014).

birth so an increase in midwifery care could reduce vaccination. In all but one state that regulated non-nursing midwives, midwives are not licensed to give this vaccination. For this reason, the first dose of Hep B vaccination could be delayed or never given. Figure A3 shows estimates of the average Hep B vaccination at the time of birth. I find no measurable effect of midwifery licensing on vaccination of Hep B at birth.

V.1.4. Maternal Health Effects

I have shown that increasing access to home birth through midwife licensing does not lead to measurable significant negative infant health effects outside some evidence of neonatal seizures. I also estimate the effects of midwifery licensing on maternal health. I do so by using HCUP hospital discharge data to estimate the effect on rates of discharges for delivery complications on the sample of women aged 15-44. Table 14 shows these estimates. While imprecise, I can rule out an increase of 53 hospitalizations per 1000 births for delivery complications. Notice that I do not study the effects on maternal mortality because the measurement of maternal mortality available in the NCHS multiple causes of death files is inaccurately reported with a 2003 change in death records (Hoyert and Miniño, 2020). For this reason, I do not include mortality as an outcome measure.

V.2. Sensitivity Checks

In this section, I test the sensitivity of results to the model choice, choices made within Callaway and Sant'Anna (2021), and sample restrictions. Firstly, I show that the estimates on home birth are robust to other estimators. Figure A4 shows event study estimates using Borusyak, Jaravel, and Spiess (2021), de Chaisemartin and D'Haultfoeuille (2019), Sun and Abraham (2021), Callaway and Sant'Anna (2021), and ordinary least squares. Regardless of the estimator, the estimates are all similar in magnitude and significance. In addition to other models, I estimate sensitivity to choices made within Callaway and Sant'Anna (2021). Firstly, for the main results, the event study estimates are calculated using a symmetric base period as is traditional in TWFE, but Callaway and

Sant'Anna (2021) also allows use of asymmetric base periods, so in Figure A5, I present results using a varying base period. As Callaway and Sant'Anna (2021) describes, these are alternative methods of reporting the same information and should not change the conclusion about whether the parallel trends assumption was violated. To address concerns of violations of parallel pre-trends, I show the robust inference and sensitivity analysis created by Rambachan and Roth (2023) in Figure A6. The main result is robust to assuming the post-treatment violation of parallel trends is no bigger than half of the worst pre-treatment violation at the 10% significance level.

Next, I show unweighted results and results where the control group consists only of "never-treated" states. Table 4 shows that estimates are robust to only using never treated states as the comparison group instead of including both not-yet and never treated. Estimates are also robust to excluding weighting by the number of births in each state and to including covariates for state level unemployment, poverty level, and median household income.

In Panel A of Table A2, I show robustness to including states in which midwives are always allowed, but there is no formal regulation, as never treated.[32] Ex-ante I expect the effect to be smaller when including places with fewer restrictions of midwives in the comparison group. Table A2 shows that, in fact, the magnitude of the effect of licensing on home birth is slightly smaller but remains statistically significant. I also show robustness to including New York and New Hampshire which license CM and no other type of non-nurse midwives in Panel B of Table A2. Panel C of Table A2 shows the robustness of the main estimates to excluding 2020 and 2021 to address concerns that COVID-19 is driving the increase in home births. Next, as Table A3, Table A4, and Table A5 show, the results are robust to Group Time Aggregation.[33]. Table A6 reports the significance for home birth heterogeneity adjusting for multiple hypothesis testing using a Bonferroni test (Bonferroni, 1935). I find that estimates for first births, college educated women, and low-risk women remain statistically significant. By contrast, the estimated effect on 5 minute APGAR score and neonatal seizures are not robust to Bonferroni multiple hypothesis testing as

[32]The states that always allow non-nurse midwives but never regulate them are HI, ME, NV, and WV. These states are dropped from the main analysis.

[33]The Callaway and Sant'Anna (2021) estimator in Stata automatically reports p-values adjusted for multiple hypothesis testing for the group aggregations (Callaway and Sant'Anna, 2023)

shown in Table A7 and Table A8.

Given that home births are a small portion of total births in the US, there is concern that I may not be able to pick up effects in health outcomes due to power limitations. To address this, I use a power analysis to determine minimum detectable (MDE) effects of midwifery licensing laws on infant health outcomes. In Tables A9 and A10 I report each MDE and the corresponding point estimate from the main specification. For outcomes, low birth weight, 5 minute APGAR score, total number of prenatal visits, premature birth, inductions, cesareans, forceps delivery, NICU admission and neonatal seizures I can reliably detect effects that are smaller than the point estimate. This suggests the true effect sizes are near zero. However, minimum detectable effects for birth weight, very low birth weight, assisted ventilation, trial of labor, and delta apgar are larger than the point estimate, meaning I lack power to rule out the estimated effects, but remain able to rule out large effects.

VI. Financial Costs of Home Birth

When discussing home births, an important consideration is the financial costs. Costs of birth differ vastly based on the place of delivery. In 2022, the average cost for vaginal delivery in hospitals was $14,768 out of pocket and $2,655 for those with insurance (KFF, 2022). On the other hand, the average cost of home birth was $4,650. While the total cost is significantly lower for home birth, most insurance companies do not cover home birth. Private insurance coverage of CPMs is mandated in 6 states, and 13 states include CPMs in their Medicaid plans. Likely due to the lack of insurance, 61 percent of all home births are paid for out of pocket, 19 percent are covered by private insurance, and 17 percent are covered by public insurance.

The increase in home births caused by midwifery licensing is driven primarily by women paying for the birth out of pocket who are positively selected. Figure 9 shows these estimates by payment type, and panel (c) particularly reports estimates for self-payment. Because I find a limited measurable effect of midwifery licensing on most measures of infant health, providing private and public insurance coverage for low-risk home births could reduce the overall costs of births without negatively impacting infant health. However, as Daysal, Trandafir, and van Ewijk (2015) find in the Netherlands where healthcare is universal, even after sorting on risk, low-income mothers have worse health outcomes when delivering at home as opposed to a hospital. Thus, instead of mandating insurance coverage of home births, if states allow people willing to pay for home births to up-take it out of pocket, it may increase access to hospital care for women who need it without negatively impacting the health of those who choose home births.

VII. Conclusion

In this research, I show that mothers respond to an expansion of birthing provider choice and healthcare access by having more home births. Particularly, when states increase access to alternative birthing options and prenatal care through licensing of non-nurse midwives, I find a 20-30 percent increase in home births. The increase is driven by positively selected mothers who are college-educated, low-risk, and pay out of pocket. This increase in home births does not correspond to a significant negative impact on infant health, aside from a possible increase in the occurrences of neonatal seizures, an extremely rare outcome. I am limited by power and data to detect impacts on some health outcomes, but remain able to detect effects for outcomes such as low birth weight, premature birth, prenatal visits, and inductions.

These results suggest that increasing access to non-traditional medical professionals like non-nurse midwives can improve the welfare of mothers by increasing their choices. However, analyses suggest that despite many states pushing for midwifery licensing to fill a gap in maternal healthcare in rural places, when midwives become available, it is not the most vulnerable mothers in those rural places who take up this alternative. My estimates suggest that midwifery legislation can explain around 20 percent of the increase in home births from 1992 to 2019.[34]

This movement to home births by low-risk women with a high willingness to pay for home birth may relieve space in hospitals for high-risk mothers who require obstetric attention. However, as

[34] In 1992, the number of home births was around 25,000, and in 2019, it was up to 45,000. Of these 20,000 additional home births annually, midwifery legislation accounts for around 4,000.

women in the US shift to using non-traditional birth settings like home births and birthing facilities the causal impact of delivery place on maternal and infant health remains an open question. Future research should directly test how home birth itself impacts health in the US.

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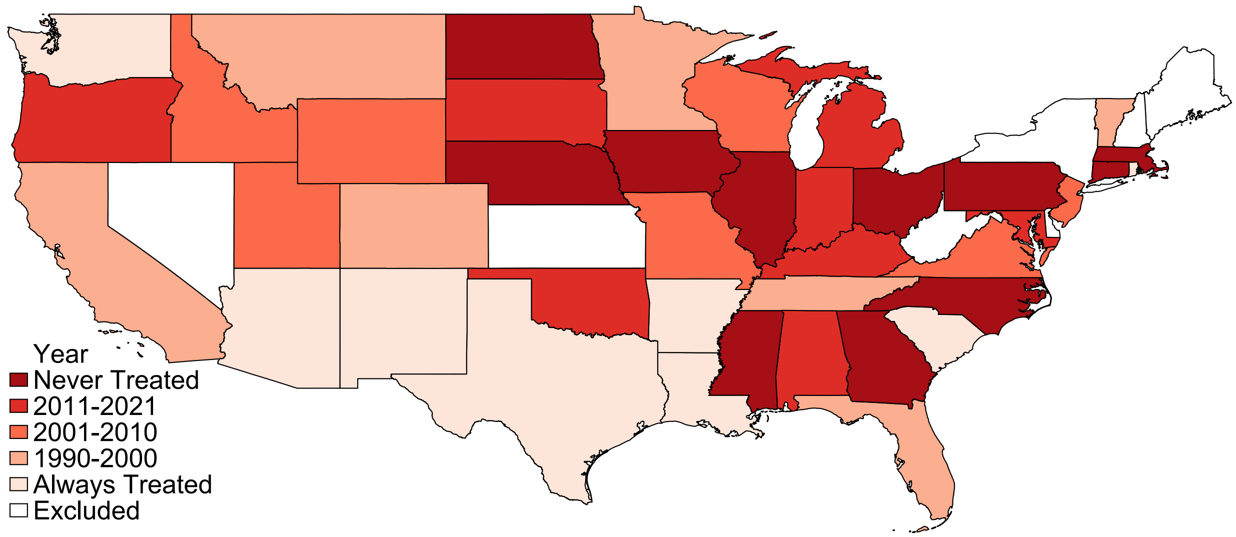
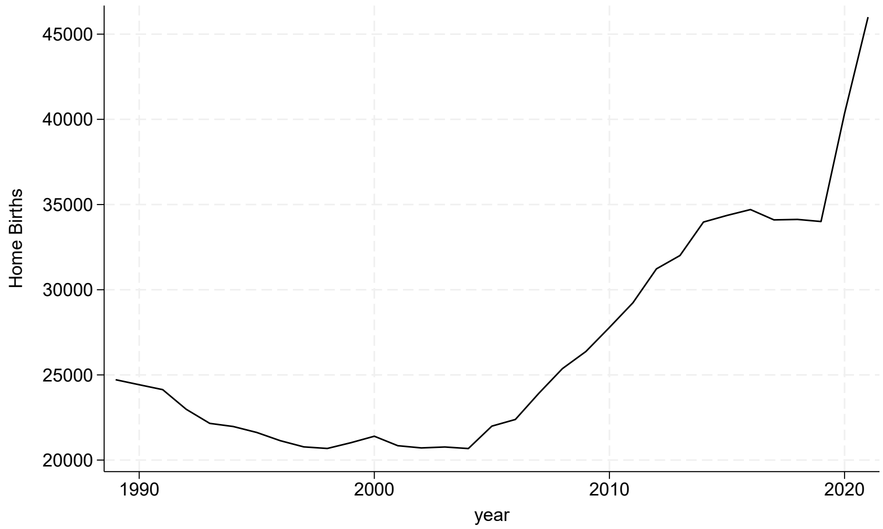
A. Figures and Tables

Figure 1 — Timing of Midwifery Licensing Legislation



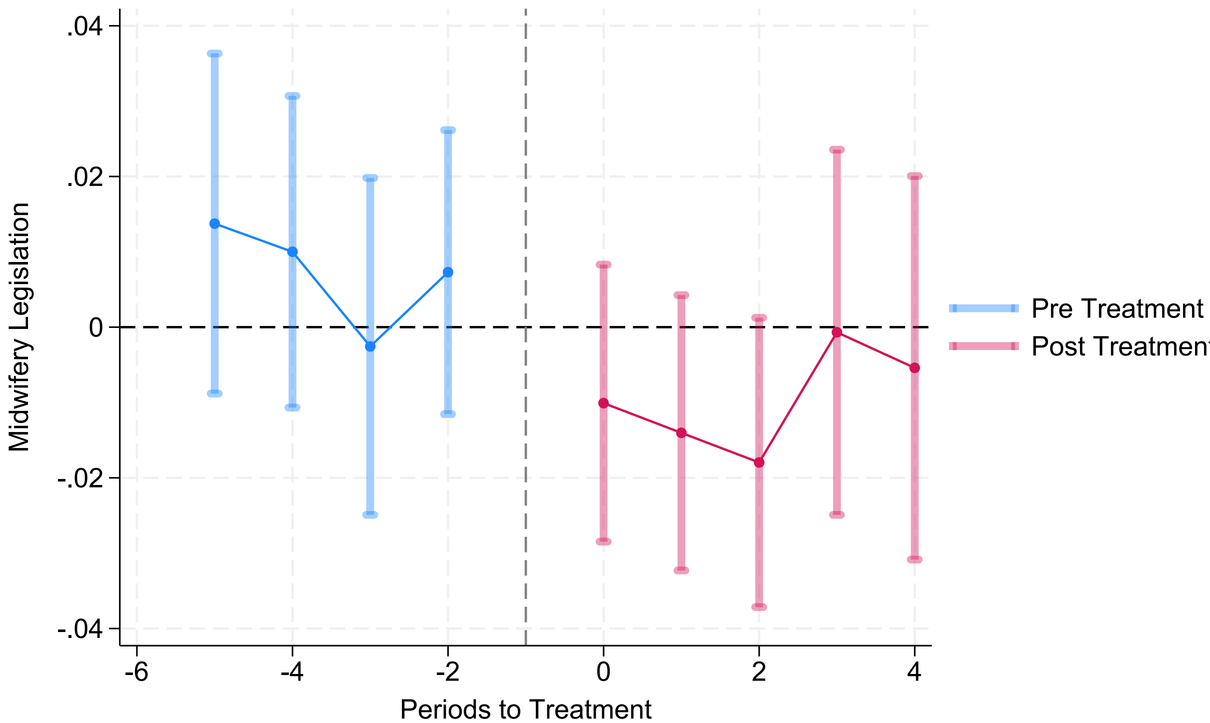
Notes: The details on state non-nursing midwifery legislation can be found in Appendix A. Never-treated states are those that have not passed laws to allow the licensing of non-nurse midwives by 2021. New York and New Hampshire are excluded because they only allow Certified Midwives. Hawaii, Kansas, Maine, Nevada and West Virginia are excluded because they view birth as natural instead of medical, and Delaware is excluded due to conflicting reports of licensing year.

Figure 2 — Number of Home Births in the U.S



Notes: Natality data is from the NCHS from 1989–2021. The Figure plots the number of births in all states from 1989 to 2021.

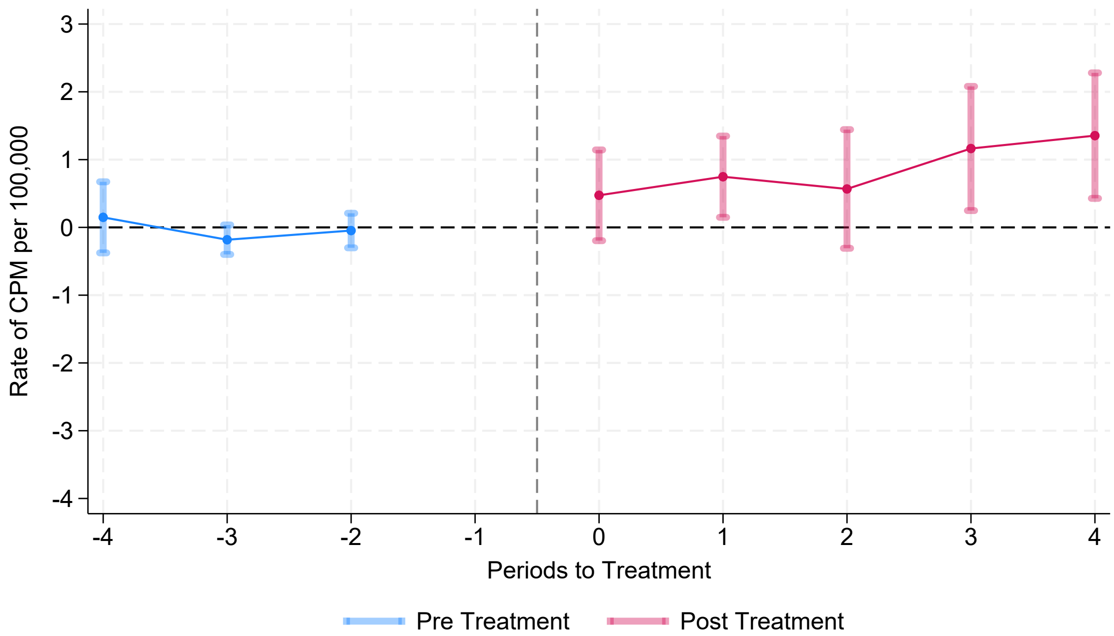
FIGURE 3 — Effect of Hospital Closures on Midwifery Licensing



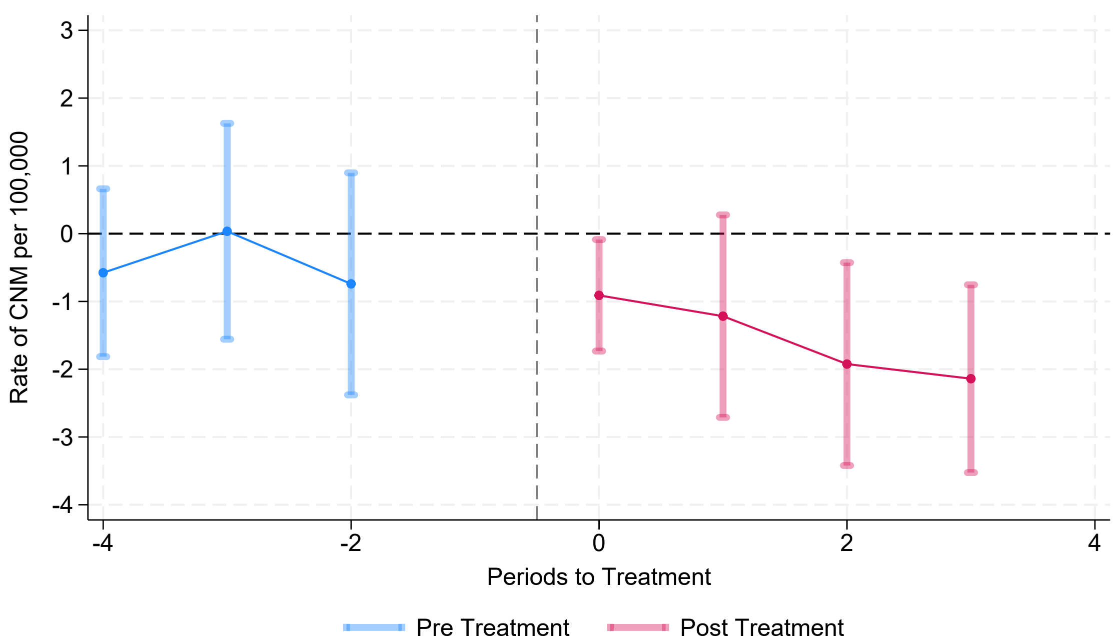
*Notes:*Natality data is from the NCHS from 1989–2021. The figure plots event estimates and their respective 95% confidence intervals from Equation (1). The treatment is the first hospital closure at the county level within a state, and the outcome of interest is if a state has midwifery licensing.

FIGURE 4 — Effects of Midwifery Licensing on Supply of Midwives

Panel A. Certified Professional Midwives

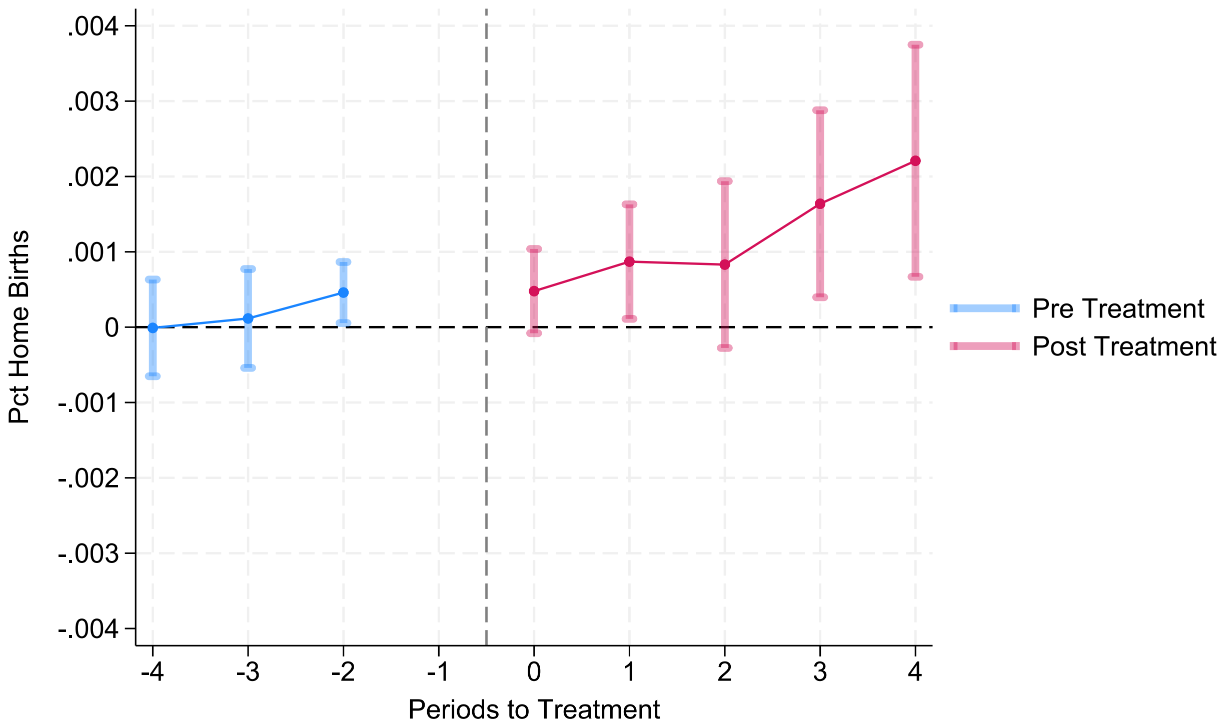


Panel B. Certified Nurse Midwives



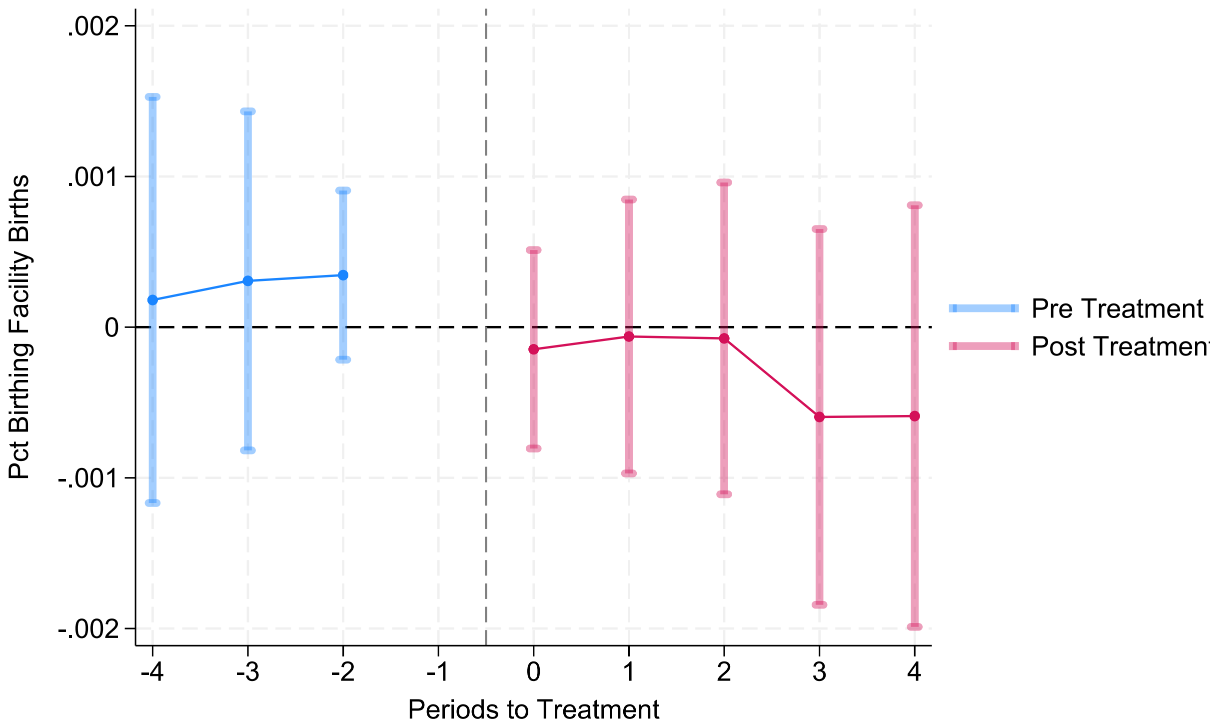
Notes: Data on the number of CNM in each state is from The Nurse Practioner's annual legislative update, which publishes counts of licensed CNM from 2000–2021. Data on the supply of CPM is from The North American Registry of Midwives (NARM) annual counts of CPM in each state from 2001–2020. This data is missing for 2011, 2017, and 2018, so results should be interpreted cautiously.The figure plots event estimates and their respective 95% confidence intervals from Equation (1).

Figure 5 — Effects of Midwifery Licensing on Percent of Home Births



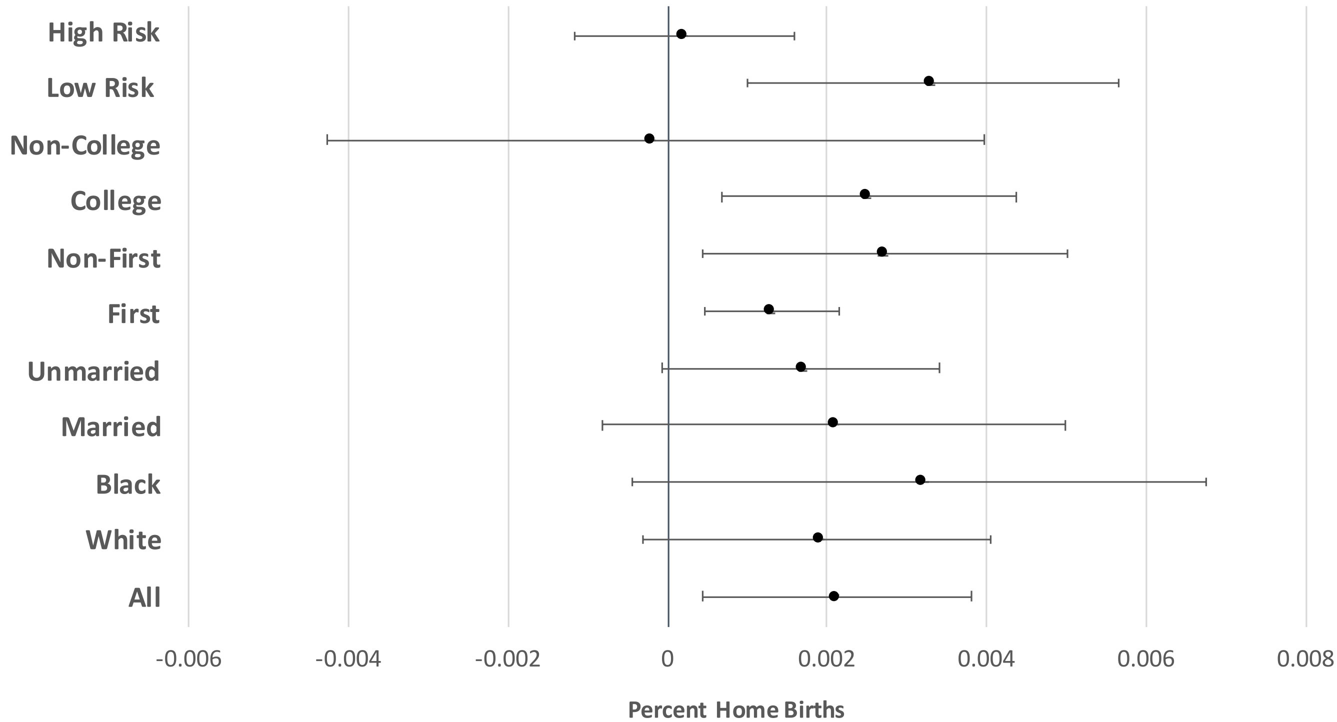
Notes: Natality data is from the NCHS from 1989–2021. The panel plots ATT event estimates and their respective 95% confidence intervals from Equation (1) where the outcome is percent of births occurring at home. The comparison group includes never treated and not yet treated states

Figure 6 — Effects of Midwifery Licensing on Percent of Births in Birthing Facilities



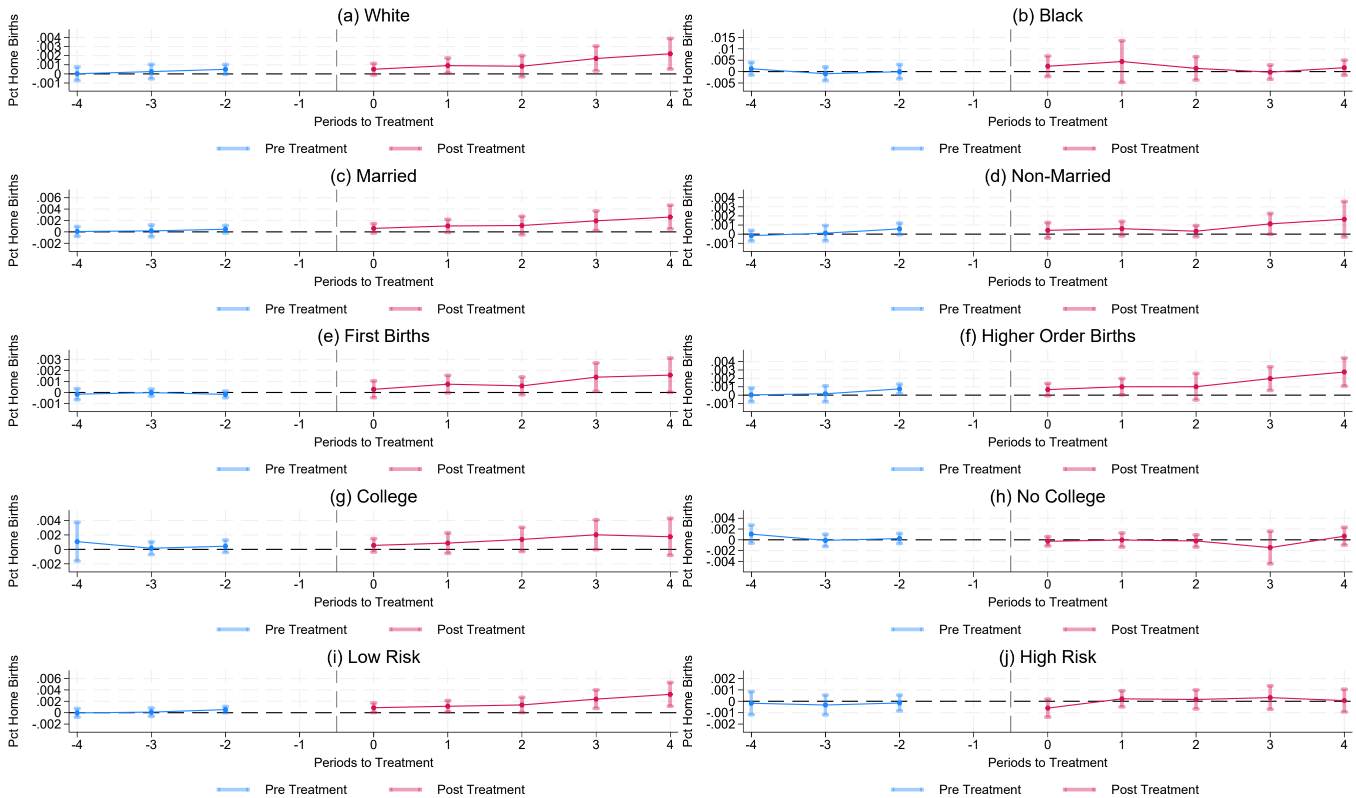
Notes: Natality data is from the NCHS from 1989–2021. The panel plots ATT event estimates and their respective 95% confidence intervals from Equation (1) where the outcome is percent of births occurring in birthing facilities. The comparison group includes never treated and not yet treated states

Figure 7 — Effects of Midwifery Licensing on Percent of Home Births - Heterogeneity



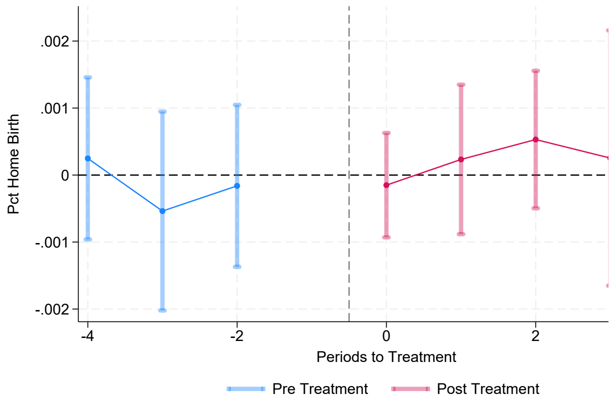
Notes: Natality data is from the NCHS from 1989–2021. Each point represents the simple ATT coefficient and their 95% confidence interval from Equation (1) for a subgroup of women. The comparison group includes never treated and not yet treated states

FIGURE 8 — Effects of Midwifery Licensing on Percent of Home Births - Heterogeneity

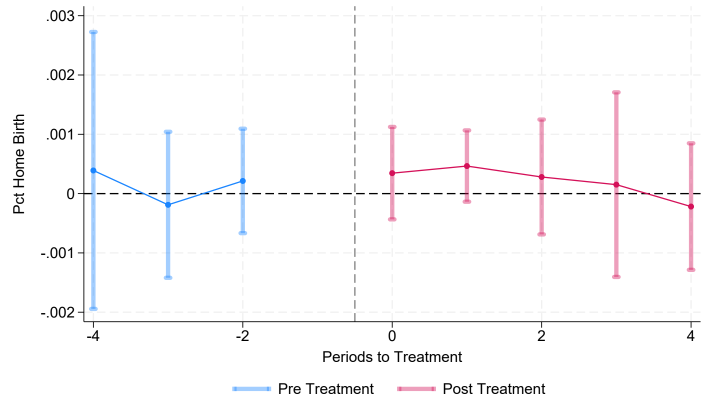


Notes: Natality data is from the NCHS from 1989–2021. The panel plots ATT event estimates and their respective 95% confidence intervals from Equation (1) where the outcome is percent of home births for a specific subroup of women. The comparison group includes never treated and not yet treated states

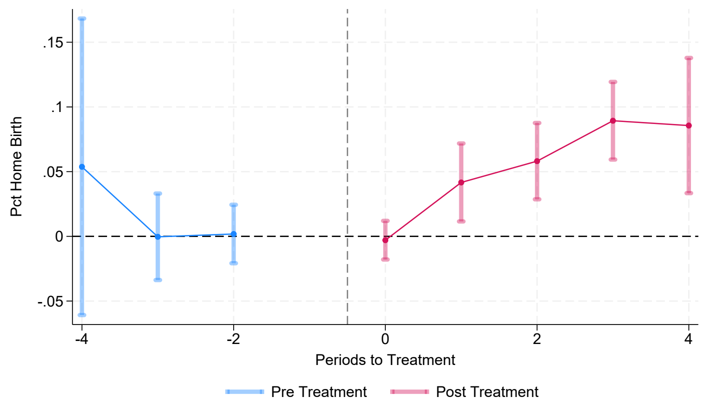
FIGURE 9 — Effects of Midwifery Licensing on Percent Home Births, by Payment Type



(A) Public Insurance



(B) Private Insurance

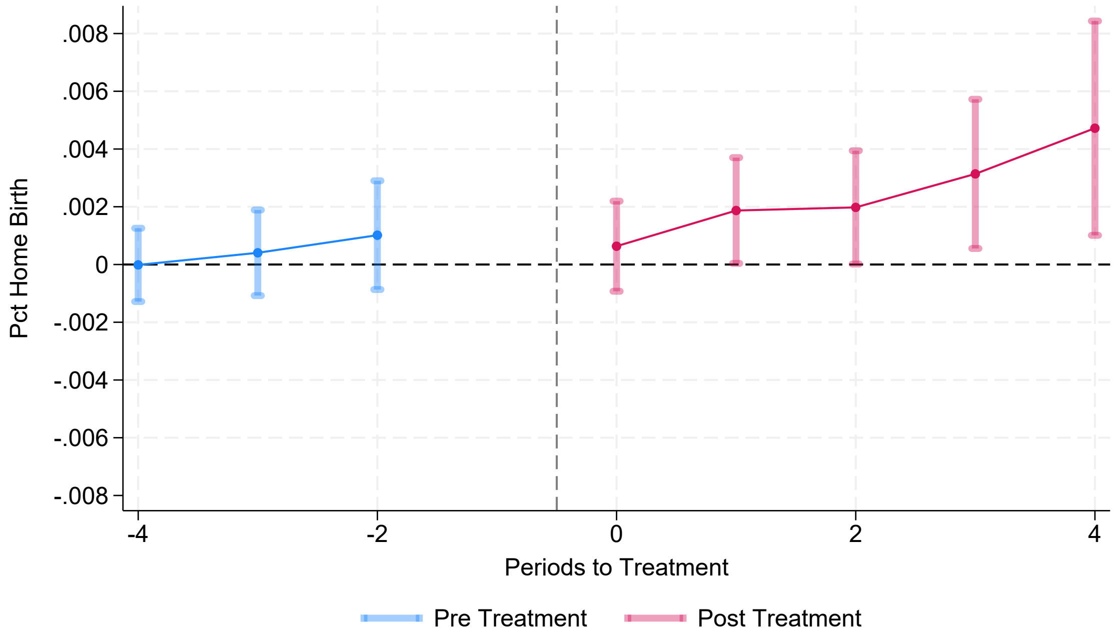


(C) Self Pay

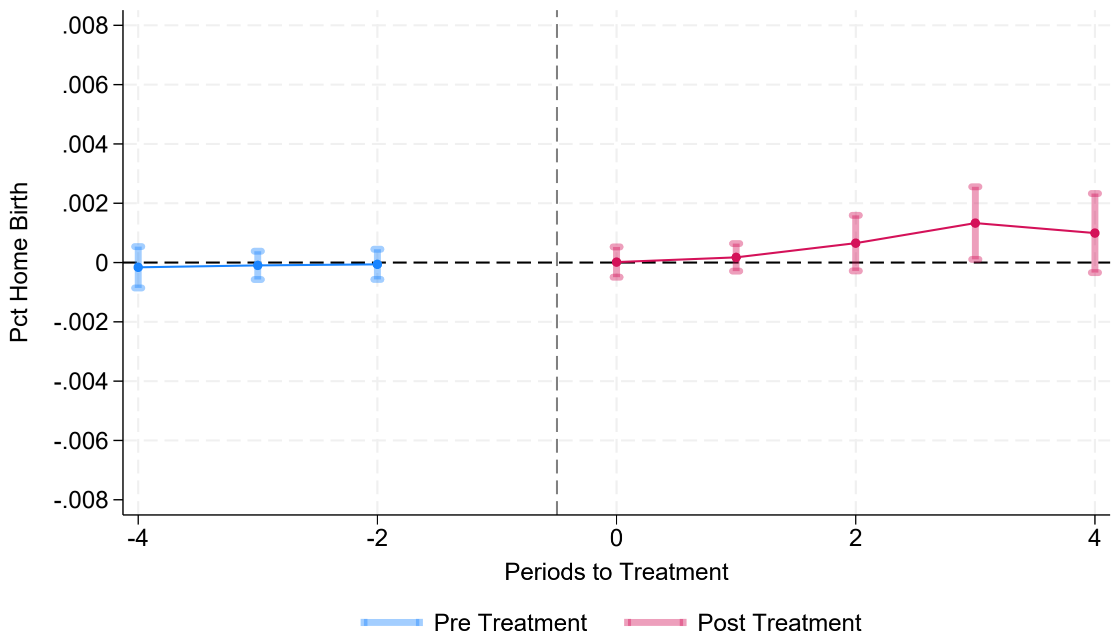
Notes: Natality data is from the NCHS from 2009-2021. The panel plots ATT event estimates and their respective 95% confidence intervals from Equation (1) where treatment is midwifery licensing. The outcome is the percentage of home births for each payment type. The comparison group includes never-treated and not-yet-treated states.

Figure 10 — Effects of Midwifery Licensing on Percent Home Births, differing county characteristics

Panel A. Non-Metro Counties

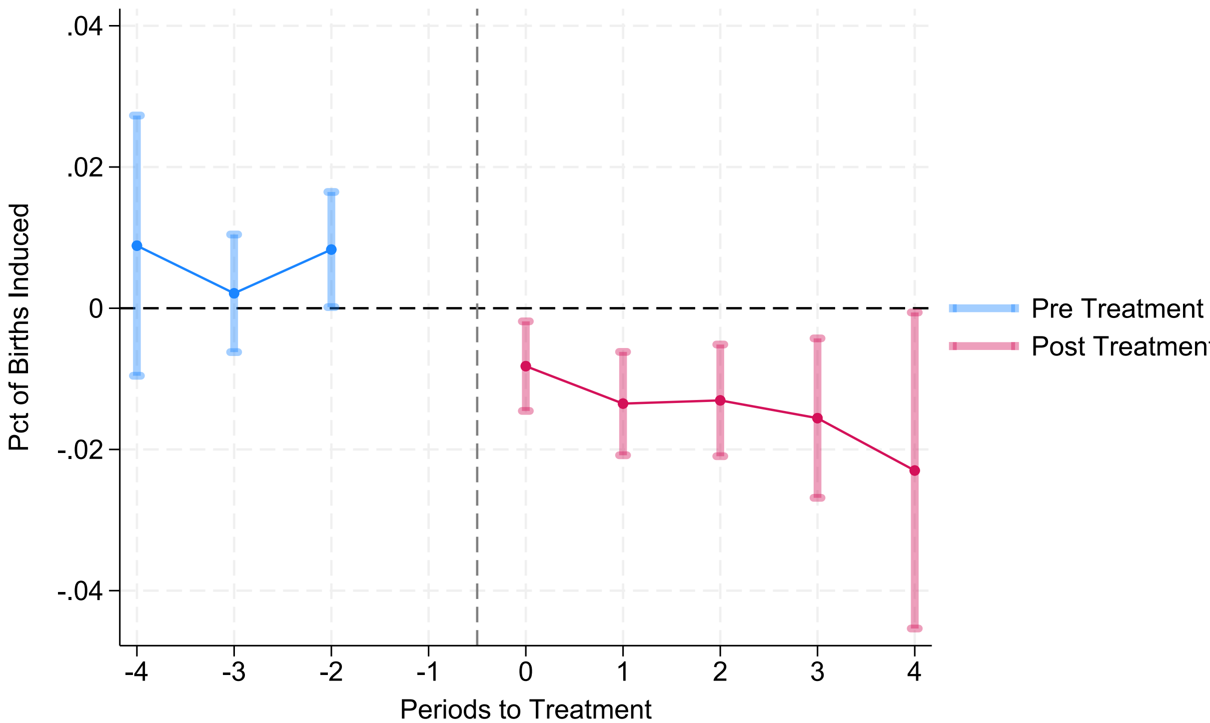


Panel B. Metro Counties



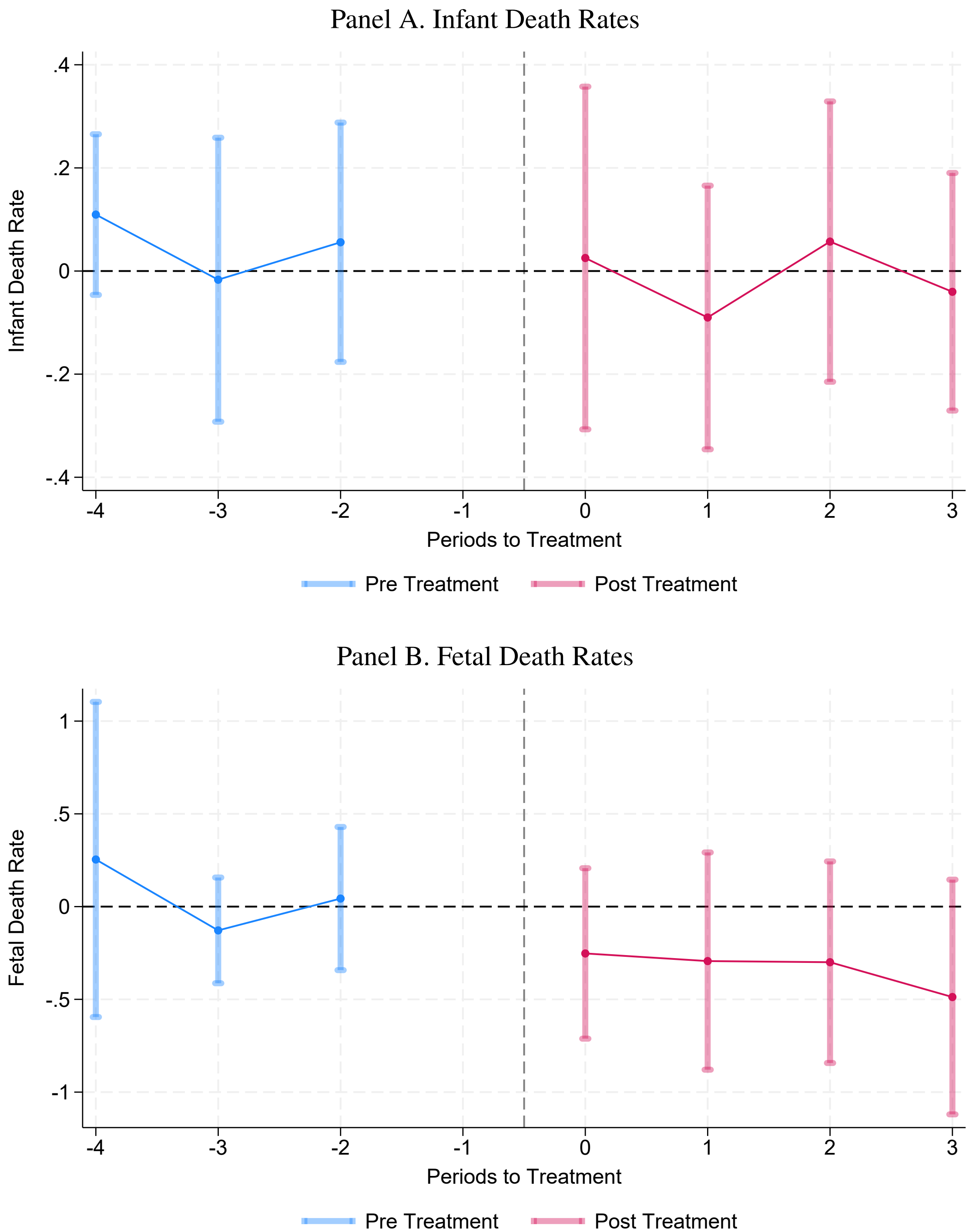
Notes: Natality data is from the NCHS from 1989–2021. The panel plots ATT event estimates and their respective 95% confidence intervals from Equation (1), where the outcome is the percent of home births. Figure (A) shows estimates for all counties defined as non-metro and Figure (B) shows estimates for counties defined as metro. The comparison group includes never treated and not-yet-treated states. The metro distinction is from the 2013 Urban-Rural Continuum from the USDA.

Figure 11 — Effects of Midwifery Licensing on Percent Induced Births



Notes: Natality data is from the NCHS from 1989–2021. The panel plots ATT event estimates and their respective 95% confidence intervals from Equation (1) where the outcome variable is percent inductions. The comparison group includes never treated and not yet treated states

Figure 12 — Effects of Midwifery Licensing on Death Rate



Notes: The figure plots ATT event estimates and their respective 95% confidence intervals from Equation (1). Infant death records are from NCHS mortality records from 1989–2021, and infant death is limited to deaths within 24 hours of birth. The death rate is per 1000 births. Fetal death records come from CDC Wonder 2005–2021. The comparison group includes never-treated and not-yet-treated states.

TABLE 1 — Descriptive Statistics for Treatment and Control States - Midwifery Licensing

	Eventually Treated States		Never Treated	
	Mean	Std. dev.	Mean	Std. dev.
	(1)		(2)	
Births per 1,000 females aged 15-44	0.072	0.009	0.066	0.004
Percent Home Births	0.006	0.003	0.006	0.002
Percent Hospital Births	0.990	0.005	0.992	0.004
Percent White	0.840	0.064	0.839	0.084
Percent Nonwhite	0.160	0.064	0.161	0.084
Percent Females aged 15-44	0.118	0.003	0.117	0.004
Median Household Income	34946.735	4842.073	34622.452	5265.178
Unemployment Rate	0.347	0.027	0.351	0.028
Below Pov level	0.135	0.032	0.125	0.036
Observations	72		44	

Notes: Individual-level natality data is from NCHS from 1989–2021. Descriptive statistics include the means and standard deviations for the listed outcomes before any treatment from 1989–1992. Columns (1) and (2) present means and standard deviations for the 18 eventually treated states for each listed outcome, and Columns (3) and (4) present means and standard deviations for the 11 never treated states.

Table 2 — Effect of Midwifery Licensing Timing on observable characteristics

	Age	Married	White	Black	First Birth	College	CNM SOP	PQC est	Med.HH Income
Midwifery Law	-0.0225	-0.0008	-0.0102	0.0095	0.0071	-0.0058	-0.0182	0.1811	1,281.2650
Std. Err.	0.0320	0.0093	0.0092	0.0084	0.0048	0.0098	0.1105	0.2153	1,329.9032
P-Value	0.4804	0.9303	0.2695	0.2621	0.1416	0.5530	0.8695	0.4003	0.3353
Mean .	4.0464	0.6724	0.8100	0.1355	0.4023	0.5339	0.1481	0.1892	51,284.8036
Observations	891	891	891	891	891	891	891	750	891

Notes: Individual-level natality data is from NCHS from 1989–2015. Each column reports estimates for a regression with a different time varying maternal or state characteristic where the treatment is midwifery licensing. CNM SOP refers to changes in the scope of practice for CNMs froom 1989 to 2015. PQC est refers to the timing of state-level initiatives called Perinatal Quality Collaboratives aimed at improving maternal and infant health.

Table 3 — Correlation Between Baseline Outcome Variables and Treatment Timing

	Treatment Timing
1992 Home Births	0.601337
	(18.1639)
1992 Inductions	-0.723249
	(2.3183)
1992 Births by Nurse Midwives	4.720821
	(3.5455)
1992 Births by Other Midwives	-0.468530
	(10.3373)
Observations	957

Notes: Individual-level natality data is from NCHS from 1989–2021. Each row reports estimates for where the baseline outcomes in 1992 are on the left-hand side and midwifery legislation timing is on the right-hand side.

Table 4 — Effect of Midwifery Licensing on Home Births

	(1)	(2)	(3)	(4)	(5)
Midwifery Law	0.0021**	0.0019***	0.0024***	0.0022**	0.0020***
Std. Err.	(0.0008)	(0.0007)	(0.0007)	(0.0009)	(0.0008)
Mean Dep. Var.	0.0067	0.0067	0.0067	0.0067	0.0067
Observations	957	957	950	957	957
Comparison Group	Not Yet & Never	Not Yet & Never	Not Yet & Never	Never	Never
Weighted	Yes	No	Yes	Yes	No
Covariates	No	No	Yes	No	No

Notes: Individual-level natality data is from NCHS from 1989–2021. Each column presents a Simple ATT estimate from Equation 1 where the outcome is percent home births. Columns (1), (2), and (3) present results where the comparison group includes both not yet treated and never treated states. Columns (4) and (5) present results where the comparison group includes only never treated states. Columns (1), (3), and (4) are weighted by number of births in a state. Column (3) includes state-level covariates of median household income, unemployment rate, and percent of the population below the poverty level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5 — Effect of Midwifery Licensing on Home Birth, Heterogeneity

	All	White	Black	Married	Unmarried	First	Non-first	College	No college	Low risk	High risk
Midwifery Law	0.0021**	0.0020*	0.0032*	0.0022	0.0018*	0.0013***	0.0027**	0.0026***	-0.0002	0.0033**	0.0002
Std. Err.	0.0008	0.0011	0.0018	0.0015	0.0009	0.0004	0.0011	0.0009	0.0021	0.0012	0.0007
Mean	0.0089	0.0097	0.0050	0.0113	0.0045	0.0044	0.0116	0.0088	0.0092	0.0110	0.0034
Observations	957	957	957	957	957	957	957	957	957	957	957

Notes: Individual-level natality data is from NCHS from 1989–2021. Each column presents Simple ATT estimates from Equation 1, where the outcome of interest is the percent of home births for a given subgroup. The comparison group includes both not yet and never treated states. Estimates are weighted by the number of births in each state.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6 — Effect of Midwifery Licensing on Home Birth, by County Metro Status

	Metro	Non-Metro
Midwifery Law	0.0014*	0.0045**
Std. Err.	0.0008	0.0018
Mean Dep. Var.	0.0057	0.0113
Observations	957	924

Notes: Individual-level natality data is from NCHS from 1989–2021. Each column presents Simple ATT estimates from Equation 1, where the outcome of interest is the percent home births. The comparison group includes both not yet and never treated states. Estimates are weighted by the number of births in each state. Column 1 reports the estimate for only metro counties, and column 2 reports the estimate for only non-metro counties. The metro distinction is from the 2013 Urban-Rural Continuum from the USDA

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7 — Effect of Midwifery Licensing on Home Births - State Heterogeneity

	Medicaid	Non Medicaid	High Maternal Mortality	Low Maternal Mortality	Strict States	Not-Strict States
Midwifery Law	0.0013	0.0007	0.0013**	0.0021	0.0027**	0.0024**
Std. Err.	0.0019	0.0004	0.0006	0.0013	0.0012	0.0010
Mean	0.0080	0.0041	0.0043	0.0081	0.0058	0.0070
Observations	617	330	363	582	747	726

Notes: Individual-level natality data is from NCHS from 1989–2021. Each column presents Simple ATT estimates from Equation 1, where the outcome of interest is the percent home births. The comparison group includes both not yet and never treated states. Estimates are weighted by the number of births in each state. Columns 1 and 2 report estimates for the subset of states with and without Medicaid coverage of home births, respectively. Columns 3 and 4 report estimates for states with high and low rates of maternal mortality. Columns 5 and 6 report estimates on home births for states with strict and nonstrict regulations of midwives and home births prior to licensing legislation.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8 — Effect of Midwifery Licensing on Delivery Outcomes

	Cesarean	Forcep	Induction	Total Births
Midwifery Law	0.003	0.001	-0.036**	-316.949
Std. Err.	0.006	0.003	0.014	3,510.319
Mean Dep. Var.	0.260	0.022	0.228	86,731.228
Observations	957	957	957	957

Notes: Individual-level natality data is from NCHS from 1989–2021. Each column presents ATT estimates from Equation 1, where the outcome of interest is a delivery outcome. The comparison group includes both not yet and never treated states. Estimates are weighted by the number of births in each state.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9 — Effect of Midwifery Licensing on Infant Health Outcomes

	Birthweight	LBW	VLBW	5 Min Apgar	Total Visits	Premature
Midwifery Law	-1.26146	0.00109	0.00044	0.04110**	0.11332	0.00184
Std. Err.	7.98638	0.00174	0.00046	0.01972	0.20960	0.00326
Mean	3,301.51432	0.07696	0.01360	8.83693	11.36574	0.11339
Observations	957	957	957	957	957	957

Notes: Individual-level natality data is from NCHS from 1989–2021. Each column presents ATT estimates from Equation 1, where the outcome of interest is an infant health outcome. The comparison group includes both not yet and never treated states. Estimates are weighted by the number of births in each state. Columns presents estimates of the average birthweight, probability of an infant born low birthweight infants (< 3,500 grams), very low birthweight (< 1,500 grams), 5 minute APGAR score, total number of prenatal visits, and premature births (<37 weeks).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 10 — Effect of Midwifery Licensing on Infant Health and Delivery Outcomes by Risk Group

	Birthweight	LBW	VLBW	5 Min Apgar	Total Visits	Cesarean	Induction	Forcep
Low Risk								
Midwifery Law	2.780	0.000	-0.000	0.039**	0.140	-0.001	-0.043***	0.002
Std. Err.	6.002	0.001	0.000	0.018	0.209	0.004	0.016	0.003
Mean	3,414.621	0.024	0.000	8.910	11.477	0.135	0.251	0.025
Observations	957	957	957	957	957	957	957	957
High Risk								
Midwifery Law	-10.427	0.004	0.002	0.042*	0.030	0.011	-0.023*	-0.001
Std. Err.	16.212	0.006	0.002	0.023	0.225	0.010	0.012	0.003
Mean	3,022.492	0.208	0.046	8.663	11.109	0.558	0.180	0.015
Observations	957	957	957	957	957	957	957	957

Notes: Individual-level natality data is from NCHS from 1989–2021. Each column presents ATT estimates from Equation 1, where the outcome of interest is an infant health outcome. The comparison group includes both not yet and never treated states. Estimates are weighted by the number of births in each state. Columns presents estimates of the average birthweight, probability of an infant born low birthweight infants (< 3,500 grams), very low birthweight (< 1,500 grams), 5 minute APGAR score, total number of prenatal visits, and premature births (<37 weeks). The top panel shows estimates for low-risk mothers and the bottom panel shows estimates for high-risk mothers.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 11 — Effect of Midwifery Licensing on Unanticipated Infant Health

	Ventilation	NICU	Seizure	Trial of Labor	Delta Apgar
Midwifery Law	0.0020	-0.0042	0.0001**	0.0074	-0.0905
Std. Err.	0.0037	0.0029	0.0000	0.0154	0.1097
Mean	0.0520	0.0904	0.0005	0.3137	2.5840
Observations	132	132	132	132	132

Notes: Individual-level natality data is from NCHS from 2010–2021 for the subset of states that revised their birth certificates by 2010. Each column presents Simple ATT estimates from Equation 1. The comparison group includes both not yet and never treated states. Estimates are weighted by the number of births in each state. Columns present estimates of assisted ventilation, NICU admission, neonatal seizure, trial of labor (planned attempt of vaginal delivery after cesarean), and delta apgar (the difference in 10 min APGAR score and 5 minute APGAR score). The only treated states are IN, MD, MI, OR, and SD.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 12 — Effect of Midwifery Licensing on Unanticipated Infant Health, by Risk Group

	Ventilation	NICU	Seizure	Trial of Labor	Delta Apgar
Low Risk					
Midwifery Law	0.0008	-0.0031*	0.0001*	0.0098	-0.0614
Std. Err.	0.0026	0.0018	0.0001	0.0132	0.1182
Mean Dep. Var.	0.0282	0.0404	0.0004	0.6229	3.2136
Observations	132	132	132	132	132
High Risk					
Midwifery Law	0.0054	-0.0058	0.0000	0.0041	-0.0944
Std. Err.	0.0055	0.0050	0.0001	0.0203	0.0858
Mean Dep. Var.	0.0996	0.1912	0.0007	0.1605	2.2015
Observations	132	132	132	132	132

Notes: Individual-level natality data is from NCHS from 2010–2021 for the subset of states that revised their birth certificates by 2010. Each column presents Simple ATT estimates from Equation 1. The comparison group includes both not yet and never treated states. Estimates are weighted by the number of births in each state. Columns present estimates of assisted ventilation, NICU admission, neonatal seizure, trial of labor (planned attempt of vaginal delivery after cesarean), and delta apgar (the difference in 10 min APGAR score and 5 minute APGAR score). The only treated states are IN, MD, MI, OR, and SD. The top panel shows estimates for low-risk mothers and the bottom panel shows estimates for high-risk mothers.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 13 — Effect of Midwifery Licensing on Infant Hospitalizations

	All Discharges	Non-hosp births	Delivery comp
Midwifery Law	-7.2283	0.0018	-0.0623
Std. Err.	5.5670	0.0118	0.0432
Mean Dep. Var.	7.0393	0.0287	0.0185
Observations	248	248	248

Notes: Individual-level natality data is from the HCUP National Inpatient Sample from 1998–2011. Each column presents Simple ATT estimates from Equation 1, where the outcome is the hospitalization discharge rates per 1000 births for infants 1-7 days old. The estimates are weighted with weights provided by HCUP. The comparison group includes both not yet and never treated states.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 14 — Effect of Midwifery Licensing on Maternal Hospitalizations

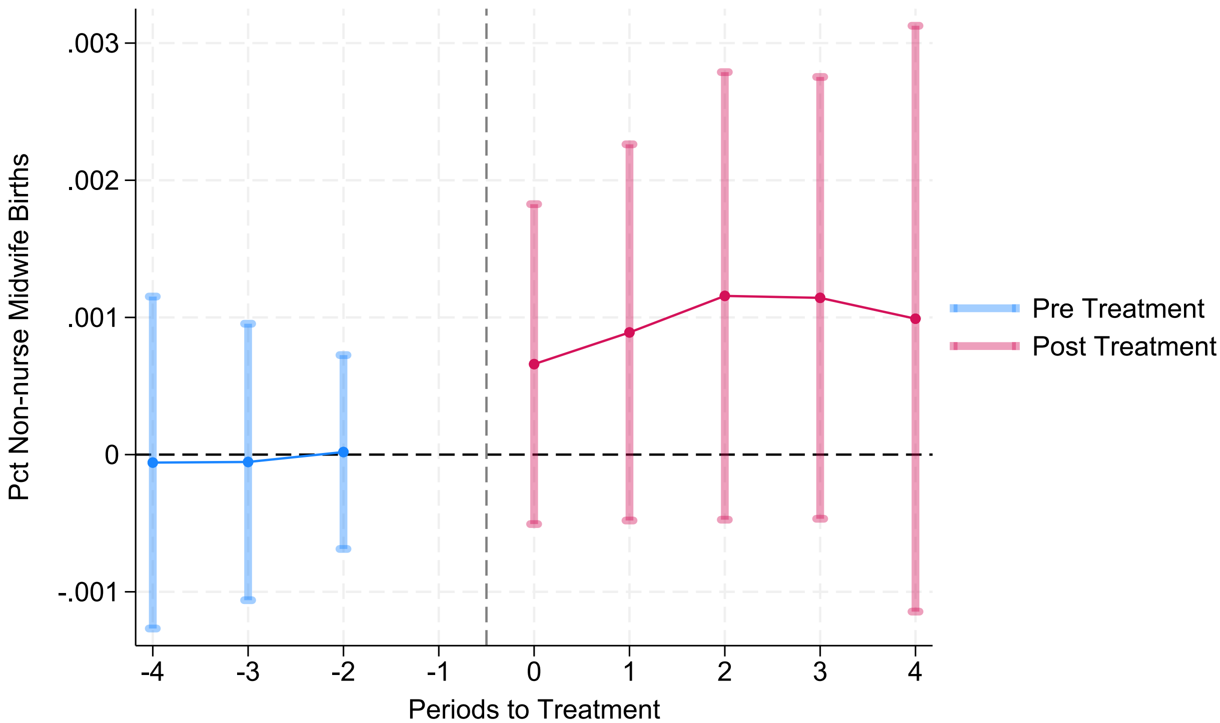
	All Discharges	Delivery comp	Hemorrhage
Midwifery Law	-528.9686	-17.2380	-1.0604
Std. Err.	568.1937	35.8592	1.8495
Mean Dep. Var.	2,265.6596	132.9106	6.0671
Observations	262	262	262

Notes: Individual-level natality data is from the HCUP National Inpatient Sample from 1998–2011. Each column presents Simple ATT estimates from Equation 1, where the outcome is the hospitalization discharge rates per 1000 births for women aged 15-44. The estimates are weighted with weights provided by HCUP. The comparison group includes both not yet and never treated states.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

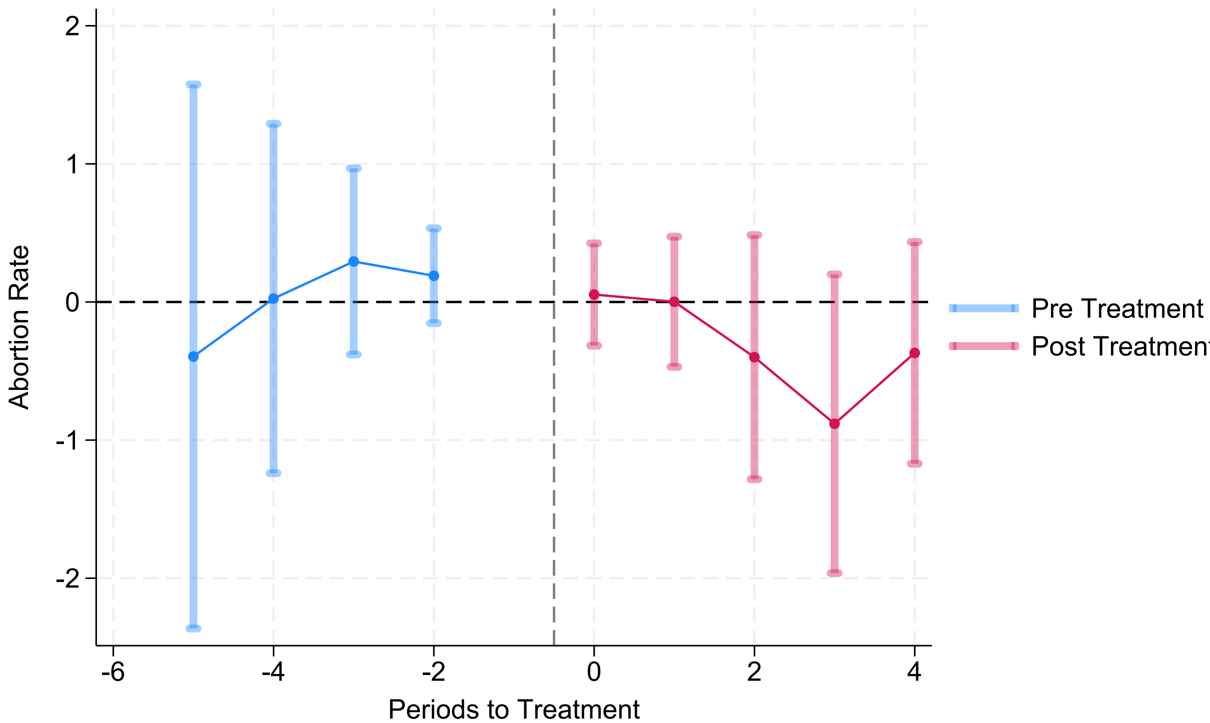
Appendix

Figure A1 — Effects of Midwifery Licensing on Percent Births by Non-Nurse Midwives



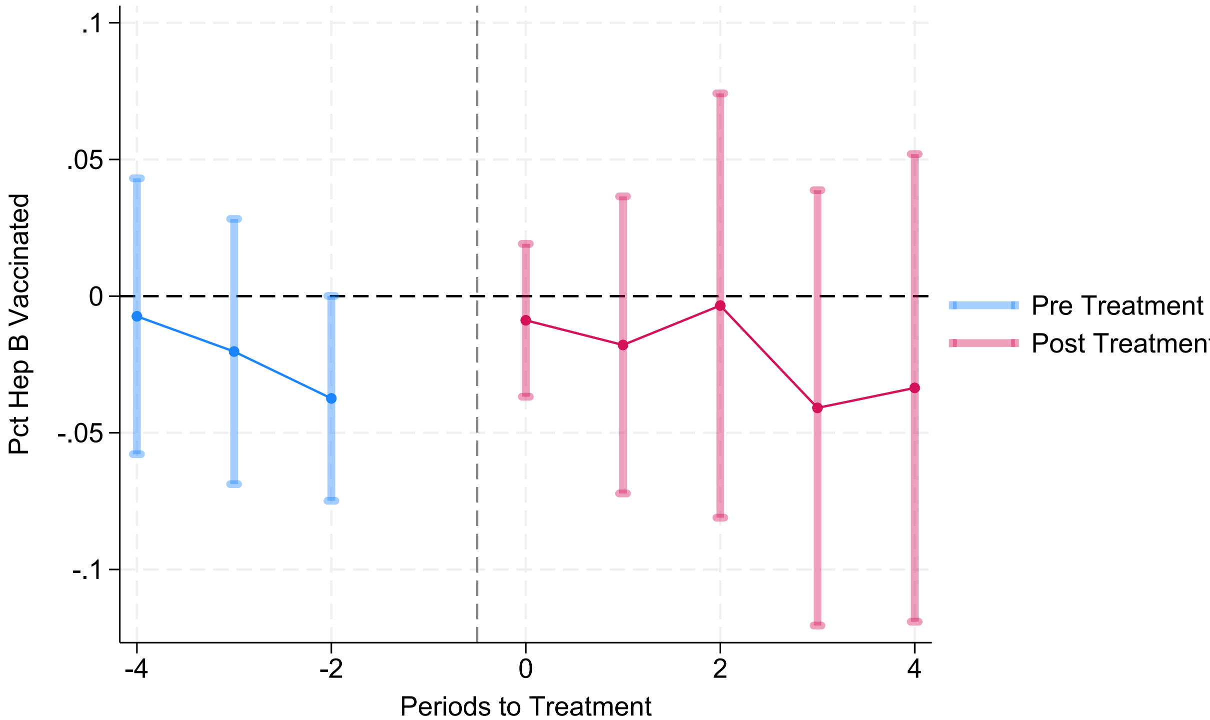
Notes: Natality data is from the NCHS from 1989–2021. The panel plots Event Study ATT estimates and their respective 95% confidence intervals from Equation (1) where the outcome is percent of births attended by a Non-nurse Midwife. The comparison group includes never treated and not yet treated states.

Figure A2 — Effects of Midwifery Licensing on Abortion



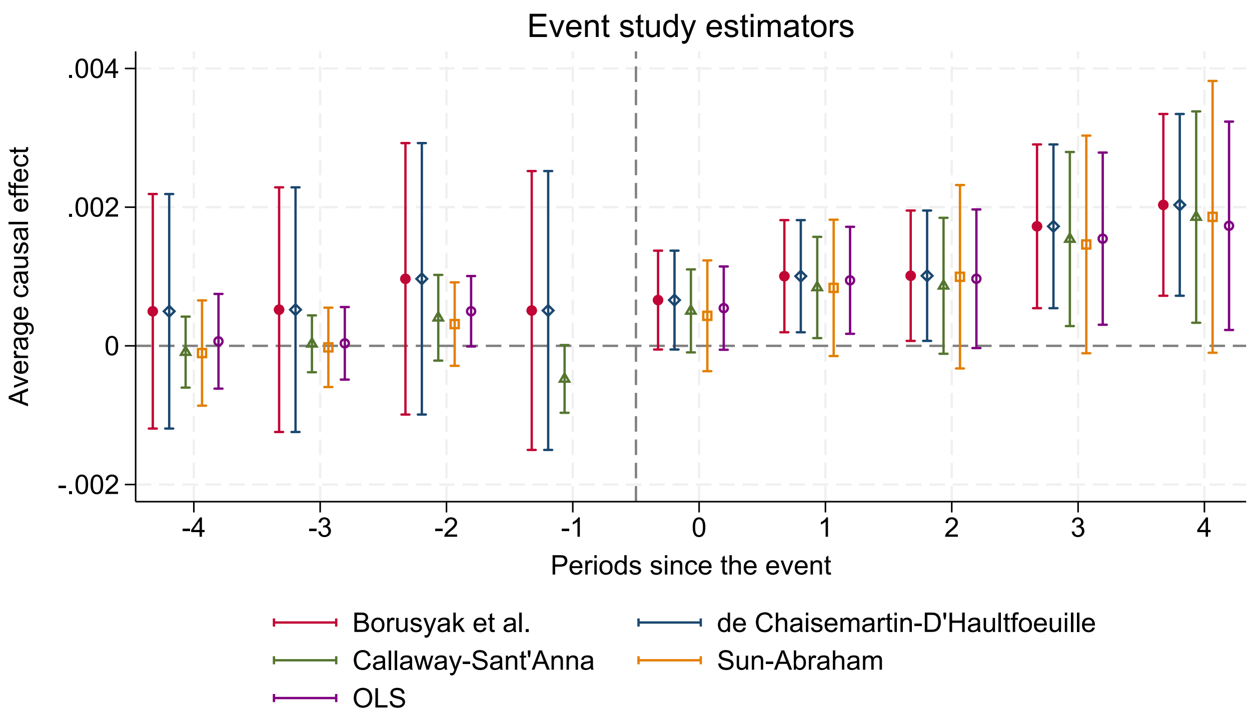
Notes: State level abortion data is from KFF from 2009–2020. The panel plots ATT event estimates and their respective 95% confidence intervals from Equation (1) where the outcome of interest is abortion rate. The comparison group includes never-treated and not-yet-treated states

Figure A3 — Effects of Midwifery Licensing on Hep B Vaccination



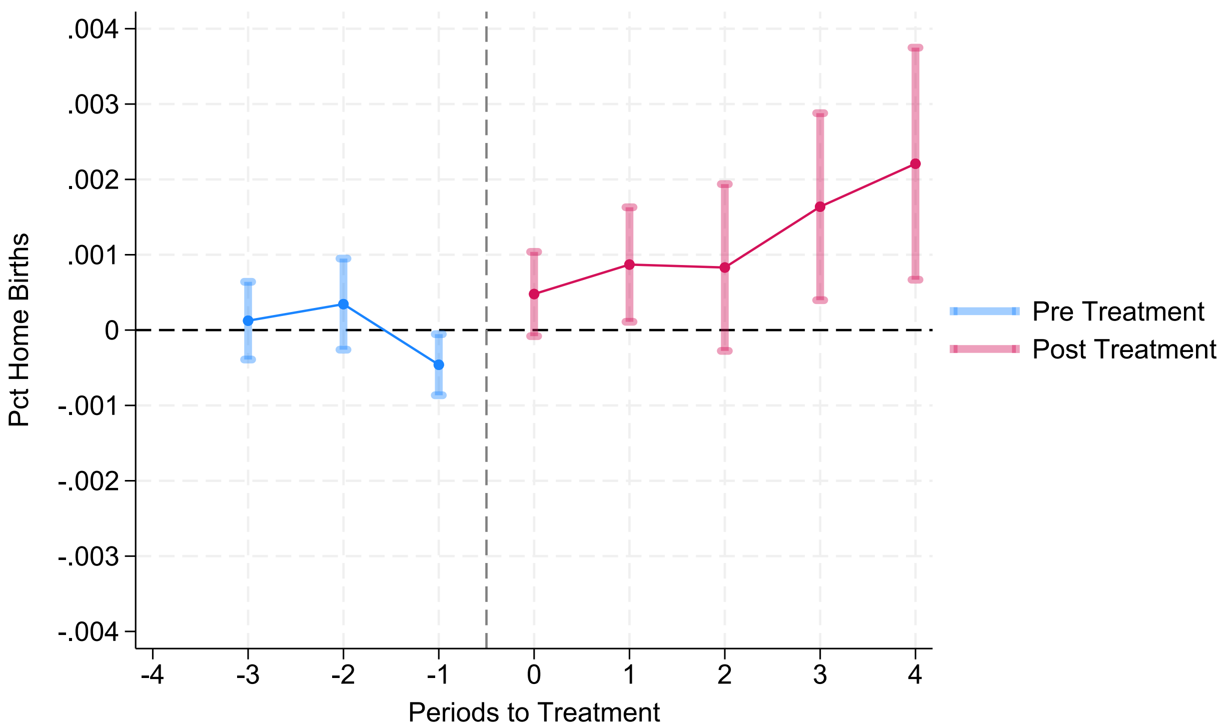
Notes: The outcome is the average hep B vaccination at birth from the National Immunization from 1992–2018. The panel plots Event Study ATT estimates and their respective 95% confidence intervals from Equation (1) where the outcome of interest is percent Hep B vaccinations. The comparison group includes never-treated and not-yet-treated states

FIGURE A4 — Effects of Midwifery Licensing on Home Birth - Five Estimators



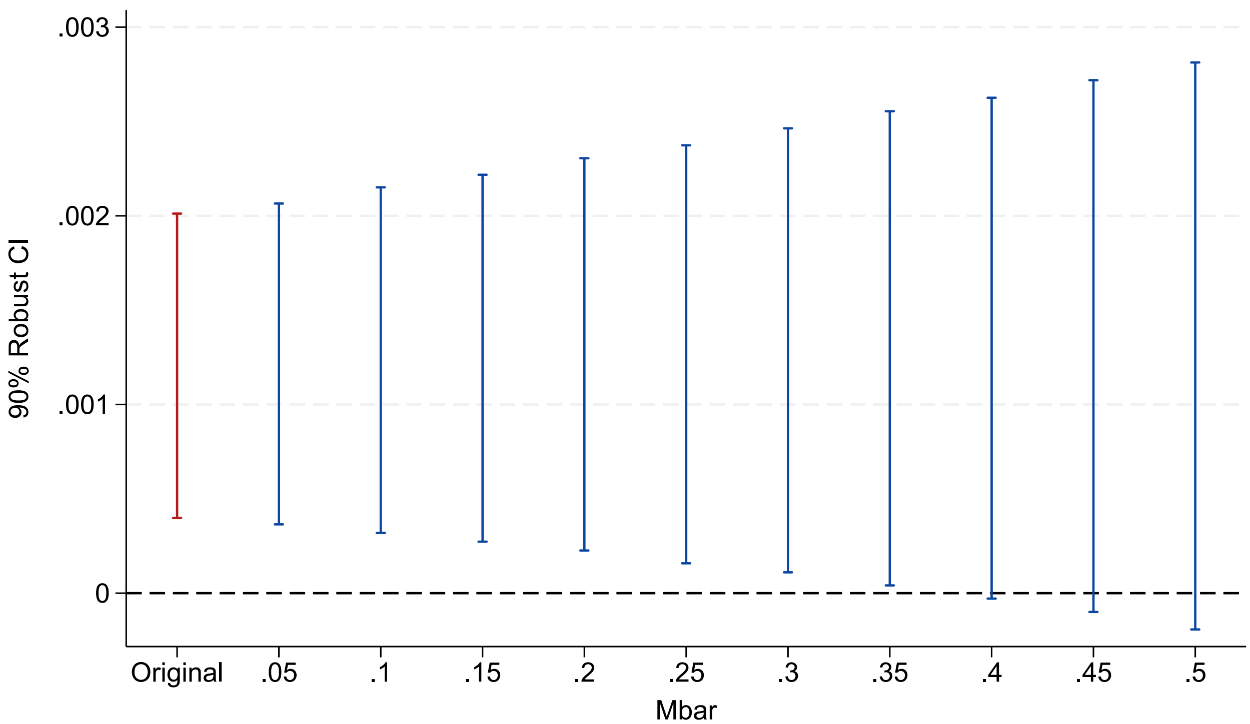
Notes: Natality data is from the NCHS from 1989–2021. The panel plots event estimates from Borusyak, Jaravel, and Spiess (2021), de Chaisemartin and D'Haultfoeuille (2019), Sun and Abraham (2021), Callaway and Sant'Anna (2021), and ordinary least squares, and their respective 95% confidence intervals. The comparison group includes never treated and not yet treated states

Figure A5 — Effects of Midwifery Licensing on Home Birth - Asymmetric Reference Period



Notes: Natality data is from the NCHS from 1989–2021. The panel plots event estimates where the base period is always the preceding year and their respective 95% confidence intervals. The comparison group includes never treated and not yet treated states.

Figure A6 — Rambachan and Roth (2023) Robust inference and Sensitivity analysis



Notes: Natality data is from the NCHS from 1989–2021. M = 0.5 represents half of the maximum violation of pretrends in the pre-period. The figure reports 90th percent confidence intervals for assuming a violation of M in the post-period.

Table A1 — Effect of Midwifery Licensing on Home Births, by Education, excluding states missing more than 5 percent

	College	Non-College
Midwifery Law	0.0028***	-0.0063
Std. Err.	0.0009	0.0052
Mean	0.0114	0.0113
Observations	504	504

Notes: Individual-level natality data is from NCHS from 1989–2021. Each column presents Simple ATT estimates from Equation 1, where the outcome of interest is the percent home birth for college and non-college educated women. Due to missingness in the education variable states with no information on education for more than 5 percent of women in any given year are excluded. The comparison group includes both not yet and never treated states. Estimates are weighted by the number of births in each state.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2 — Effect of Midwifery Licensing on Home Births - Sample Sensitivity

	(1)	(2)	(3)	(4)
Panel A				
Midwifery Law	0.0017**	0.0014*	0.0017**	0.0014*
Std. Err.	0.0008	0.0008	0.0009	0.0008
Mean	0.0069	0.0069	0.0069	0.0069
Observations	1,122	1,122	1,122	1,122
Panel B				
Midwifery Law	0.0020**	0.0019***	0.0020**	0.0020***
Std. Err.	0.0008	0.0007	0.0008	0.0007
Mean	0.0068	0.0068	0.0068	0.0068
Observations	990	990	990	990
Panel C				
Midwifery Law	0.0018**	0.0016**	0.0020***	0.0018**
Std. Err.	0.0008	0.0007	0.0007	0.0008
Mean	0.0067	0.0067	0.0067	0.0067
Observations	899	899	894	899
Comparison Group	Not Yet & Never	Not Yet & Never	Never	Never
Weighted	Yes	No	Yes	No

*Notes:*Individual-level natality data is from NCHS from 1989–2021. Each column presents Simple ATT estimates from Equation 1. Columns (1) and (2) present results where the comparison group includes both net yet treated and never treated states. Columns (3) and (4) present results where the comparison group includes only never treated states. Columns (1) and (3) are weighted by number of births in a state. Panel A shows estimates when including HI, ME, NV, and WV, which all allow non-nurse midwives to practice without licenses and have no regulations, as never treated states. Panel B shows estimates when including NY and NH, which only license CM as treated states. Panel C shows estimates from the main specification excluding 2020 and 2021 to address concerns over the inclusion of years during the COVID 19 pandemic.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3 — Effect of Midwifery Licensing on Home Birth, Heterogeneity - Group Time Aggregation

	All	White	Black	Married	Unmarried	First	Non-first	College	No college	Low Risk	High Risk
Midwifery Law	0.0021***	0.0021***	0.0028***	0.0025***	0.0013***	0.0011***	0.0028***	0.0023***	0.0007	0.0033***	0.0003
Std. Err.	0.0006	0.0007	0.0011	0.0009	0.0004	0.0003	0.0007	0.0006	0.0011	0.0008	0.0004
Mean	0.0089	0.0097	0.0050	0.0113	0.0045	0.0044	0.0116	0.0088	0.0092	0.0110	0.0034
Observations	957	957	957	957	957	957	957	957	957	957	957

Notes: Individual-level natality data is from NCHS from 1989–2021. Each column presents Group Time ATT estimates from Equation 1 where the outcome of interest is the percentage of home births for a given subgroup. Group ATT estimates adjust for multiple hypothesis testing. The comparison group includes both not yet and never treated states. Estimates are weighted by the number of births in each state.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4 — Effect of Midwifery Licensing on Infant Health Outcomes - Group Time Aggregation

	Birthweight	LBW	VLBW	5 Min Apgar	Total Visits	Premature
Midwifery Law	-1.19142	0.00097	0.00039	0.03642**	0.10598	0.00124
Std. Err.	4.04607	0.00094	0.00026	0.01743	0.09001	0.00158
Mean	3,301.51432	0.07696	0.01360	8.83693	11.36574	0.11339
Observations	957	957	957	957	957	957

Notes: Individual-level natality data is from NCHS from 1989–2021. Each column presents Group Time ATT estimates from Equation 1, where the outcome of interest is an infant health outcome. Group ATT estimates adjust for multiple hypothesis testing. The comparison group includes both not yet and never treated states. Estimates are weighted by the number of births in each state.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A5 — Effect of Midwifery Licensing on Delivery Outcomes - Group Time Aggregation

	Cesarean	Forcep	Induction	Total Births
Midwifery Law	0.001	0.001	-0.028***	171.981
Std. Err.	0.003	0.001	0.007	1,837.829
Mean Dep. Var.	0.260	0.022	0.228	86,731.228
Observations	957	957	957	957

Notes: Individual-level natality data is from NCHS from 1989–2021. Each column presents Group Time ATT estimates from Equation 1, where the outcome of interest is a delivery outcome. Group ATT estimates adjust for multiple hypothesis testing. The comparison group includes both not yet and never treated states. Estimates are weighted by the number of births in each state.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6 — Effect of Midwifery Licensing on Delivery Home Birth, Heterogeneity - Adjusted for Multiple Hypothesis Testing

	White	Black	Married	Unmarried	First	Non-first	College	No college	Low risk	High risk
Midwifery Law	0.0020	0.0032	0.0022	0.0018	0.0013**	0.0027	0.0026*	-0.0002	0.0033*	0.0002
Std. Error	0.0011	0.0018	0.0015	0.0009	0.0004	0.0011	0.0009	0.0021	0.0012	0.0007
Unadjusted p-value	0.07	0.08	0.13	0.05	0.00	0.01	0.01	0.92	0.00	0.74
Bonferroni p-value	0.68	0.77	1.00	0.50	0.03	0.15	0.06	1.00	0.05	1.00

Notes: Individual-level natality data is from NCHS from 1989–2021. Each column presents simple ATT estimates from Equation 1 where the outcome of interest is home births for a subgroup. The comparison group includes both not yet and never treated states. Estimates are weighted by the number of births in each state. To determine the Bonferroni adjusted p-value we multiple the unadjusted p-value by the number of tests, in this case 10, with the maximum value set at 1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A7 — Effect of Midwifery Licensing on Health and Delivery Outcomes - Adjusted for multiple hypothesis Testing

	Birthweight	LBW	VLBW	5 Min Apgar	Total Visits	Premature	Cesarean	Induction	Forcep
Midwifery Law	-1.2615	0.0011	0.0004	0.0411	0.1133	0.0018	0.0031	-0.0361*	0.0014
Std. Error	7.9864	0.0017	0.0005	0.0197	0.2096	0.0033	0.0056	0.0138	0.0032
Unadjusted p-value	0.87	0.53	0.34	0.04	0.59	0.57	0.58	0.01	0.66
Bonferroni p-value	1.00	1.00	1.00	0.33	1.00	1.00	1.00	0.08	1.00

Notes: Individual-level natality data is from NCHS from 1989–2021. Each column presents Simple ATT estimates from Equation 1, where the outcome of interest is an infant health outcome. The comparison group includes both not yet and never treated states. Estimates are weighted by the number of births in each state. To determine the Bonferroni adjusted p-value we multiple the unadjusted p-value by the number of tests, in this case 9, with the maximum value set at 1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE A8 — Effect of Midwifery Licensing on Unanticipated Infant Health - Adjusted for Multiple Hypothesis Testing

	Ventilation	NICU	Seizure	Trial of Labor	Delta Apgar
Midwifery Law	0.0020	-0.0042	0.0001	0.0074	-0.0905
Std. Error	0.0037	0.0029	0.0000	0.0154	0.1097
Unadjusted p-value	0.58	0.14	0.02	0.63	0.41
Bonferroni p-value	1.00	0.71	0.10	1.00	1.00

Notes: Individual-level natality data is from NCHS from 2010–2021 for the subset of states that revised their birth certificates by 2010. Each column presents Simple ATT estimates from Equation 1. The comparison group includes both not yet and never treated states. Estimates are weighted by the number of births in each state. The only treated states are IN, MD, MI, OR, and SD. Group ATT estimates adjust for multiple hypothesis testing. To determine the Bonferroni adjusted p-value we multiple the unadjusted p-value by the number of tests, in this case 5, with the maximum value set at one

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9 — Power Calculation for Infant Health and Delivery Outcomes

	BW	LBW	VLWB	5 Min Apgar	Total Visits	Premature	Induction	Cesarean	Forceps
MDE	1.6287	0.0005	0.0002	0.0124	0.0674	0.0008	0.0050	0.0018	0.0011
Estimated effect	-1.26146	0.00109	0.00044	0.04110*	0.11332	0.00184	-0.036***	0.003	0.002
Observations	957	957	957	957	957	957	957	957	957

Notes: Individual-level natality data is from NCHS from 1989–2021. Each column presents the minimum detectable effect (MDE) given 957 units, 28 percent treated, and 4 pre/post periods. In the second row, point estimates for each outcome is reported.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A10 — Power Calculation for Unanticipated Infant Health Outcomes

	Ventilation	NICU	Seizure	Trial of Labor	Delta Apgar
MDE	0.0038	0.0037	0.0001	0.0147	0.1061
Estimated effect	0.0020	-0.0042	0.0001*	0.0074	-0.0905
Observations	132	132	132	132	132

Notes: Individual-level natality data is from NCHS from 2010–2021 for the subset of states that revised their birth certificates by 2010. Each column presents Simple ATT estimates from Equation 1. The only treated states are IN, MD, MI, OR, and SD. Each column presents the minimum detectable effect (MDE) given 132 units, 27 percent treated, and 4 pre/post periods. In the second row, point estimates for each outcome is reported.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

State	Type of Non-nursing Midwife	Year of passage	Source
Alabama	CPM	2017	Code of Ala. § 34-19-15(c)-(d) (2017).
Arizona	LM	1957	Laws 1957, Chapter 45
Arkansas	Direct Entry	1983, 1987	Act 838 of 1983
Alaska	Direct Entry	1992	Alaska Stat. § 08.65.050 (1992).
California	LM	1993	Licensed Midwifery Practice Act of 1993
Colorado	LM	1993	H.B. 1051, 59th Gen. Assemb., Reg. Sess. (Co. 1993)
Connecticut	None		
Delaware	CM, CPM	1978, 2001-2015, 2015	24 Del. C. § 1799FF(2)-(3) (2016).
Florida	LM	1992	FL ST §§ 467.001 et seq.
Georgia	None		Ga. Admin. R. 511-5-1-.02
Hawaii			
Idaho	LM	2009	House Bill 185 An Act Relating to Midwifery
Illinois	CPM	2023	Hundsdorfer, 2021
Indiana	Direct Entry	2013	IN ST 25-23.4-1-1 et seq.
Iowa	CPM	2023	House File 265
Kansas		drop	
Kentucky	CPM	2020	Walker, 2019
Louisiana	LM	1985	Acts 1984, No. 688, §1, eff. Jan. 1, 1985.
Maine	CPM	drop	Midwives of Maine
Maryland	CPM, Direct Entry	2015	MD HEALTH OCCUP §§ 8-6C-01 et seq.
Massachusetts	None		
Michigan	CPM	2016 passed	Section 33.17101
Minnesota	CPM, Direct Entry	1999 passed	Minn. Stat. Ann. § 147D.17
Mississippi	none		Miss. Code Ann. § 73-25-33
Missouri	LM	2008	Wolff, 2020
Montana	Direct Entry	1991	Laws 1991, ch. 550, § 1.
Nebraska	None		
Nevada		Drop	Assembly Bill 386 (in consideration)
New Hampshire	CM	1999	Section 326:D
New Jersey	CM, CPM	2010	2A:53A-26
New Mexico	LM	1980	N.M. Admin. Code Ch 11, Pt 3
New York	CM	1992	Article 140 of the Education Law, §6955(2)(b) and (c)
North Carolina	none		
North Dakota	none		
Ohio	none		Athens, 2021
Oklahoma	Direct Entry, CM	2021	Senate Bill 1823
Oregon	Direct Entry	2013	OR ST §§ 687.405 et seq.
Pennsylvania	none		
Rhode Island	CM, CPM	1978	RI ST § 23-13-9
South Carolina	Direct Entry	1987	SECTION 44-89-10
South Dakota	CPM	2017	36-9C-13

Tennessee	CPM	2000	Title 36 Ch 29
Texas	Direct Entry	1983	SB 238, 68th R.S.
Utah	Direct Entry	2005	Title 58 chapter 77
Vermont	Direct Entry	1999	VT ST T. 26 §§ 4181
Virginia	LM	2005	54.1 -2957.9
Washington	Direct Entry	1980/1981	RCW 18.50.010
West Virginia		Drop	
Wisconsin	CPM	2005	Wis. Stat. § 440.982
Wyoming	CPM	2010	Wyo. Stat. § 33-46-103 (2010).

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